

Religious Frictions on the Capital Markets

Victor Andrée

22584@student.hhs.se

Katarina Warg

22555@student.hhs.se

Stockholm School of Economics
Bachelor Thesis in Finance
Spring 2014

Abstract

The Malaysian government strives to establish an Islamic capital market in synchrony with the conventional market. Looking at conventional bonds and sukuk (Islamic bonds) issued by the same entity (Government of Malaysia), denominated in the same currency and traded on the same trading platform we find a statistically significant price deviation for short and medium term maturities. We construct and compare zero yield curves for both markets as well as use the conventional yield curve to estimate the issue specific price deviation of a sukuk. This paper argues that the frictions in part can be explained by illiquidity issues and finds that the proportions of zero returns has some explanatory power on the yield spread.

Keywords: Islamic finance, sukuk, Malaysian treasury market, yield spread, illiquidity.

Tutor: Paolo Sodini

Acknowledgments: We would like to thank our tutor Paolo Sodini for valuable input and guidance.

Contents

1	Introduction	2
1.1	Outline of paper	3
2	Background	4
2.1	Basics of Islamic finance	4
2.2	The Malaysian bond market	5
3	Literature review	8
3.1	Previous research on Islamic Finance	8
3.2	Studies on illiquidity and bond yield spreads	9
4	Data	10
4.1	Market quotes: Discount, Clean Price and Bond Equivalent Yield	11
5	Theory and Methodology	12
5.1	Hypothesis development	12
5.2	Bond valuation, absence of arbitrage and the yield curve	13
5.3	Modelling the yield curve	14
5.4	Determining the yield spread	16
5.5	Liquidity and frictions	17
5.6	Choice of liquidity measure	19
6	Results	21
6.1	Yield curves	21
6.2	Model validation using arbitrage-free yields	29
6.3	Arbitrage-free yields	32
6.4	Liquidity regression results	34
7	Conclusion	39
8	Limitations and suggestions for future research	40
8.1	Limitations	40
8.2	Suggestions for future research	40
	Appendices	44
	Appendix A Sukuk principles	44
	Appendix B Complementing tables and figures using Nelson-Siegel	45

1 Introduction

On the Malaysian capital markets the conventional and Islamic financial sectors operate in parallel with each other. According to the principles of Shariah, the charging of interest is prohibited which has led to numerous financial innovations. The establishment of an Islamic market and its historical tremendous growth could potentially drive financial inclusion in a country with a majority of Muslims but can also be a cause of market frictions. As outlined in the Financial Sector Masterplan (2001) and the Financial Sector Blueprint (2011), the government strives to enhance the Islamic financial sector, aiming to establish two separate markets in synchrony.

While conventional principles still constitute the basis for policy-making and monetary policy (Iqbal & Mirakhor, 2013), the Malaysian government issues bonds on both the conventional and Islamic capital markets. The presence of two parallel markets, with similar products and cross-market investors, constitutes an interesting research topic from a no-arbitrage perspective.

Sukuk (Islamic bonds) and conventional bonds are issued by the same entity, are denominated in the same currency and traded on the same trading platform. Investors can participate in both markets and no preferential tax treatment is currently applied to either market. Some differences however prevail. The sukuk market is significantly smaller and short-selling is not allowed. Short-selling of conventional bonds is possible though heavily regulated. The Islamic derivatives market is practically non-existent.

According to the central bank of Malaysia, Bank Negara Malaysia (BNM, 2014), the structure at issuance is different for a sukuk. A sukuk is always based on the transfer of an asset and cash flows are created as a result of this transaction. However, the asset only forms part of the initial issuance and does not constitute collateral. Like government issued conventional bonds, sukuk have a fixed maturity, can bear coupons and the conventional yield-price relationship prevails. While some sukuk structures applied on the corporate bond market can differ from conventional principles in terms of coupon payments, government issues all have predetermined coupons. In effect, once the sukuk has reached the secondary market issuance structures are irrelevant. In this paper, we investigate the differences between sukuk and conventional bonds based on two hypotheses:

Hypothesis 1: *The yield of a sukuk should be equal to or lower than the yield of a comparable conventional bond.*

Hypothesis 2: *Illiquidity issues can cause a positive yield spread between Islamic and conventional bonds.*

In the absence of arbitrage opportunities, a sukuk and a conventional bond issued by the government of Malaysia with matching cash flows should have the same fundamental value. A conventional investor should be indifferent between the two types while a Shariah conscious investor should prefer sukuk. The Shariah conscious investor base should therefore accept a lower yield,

driving up the prices, as argued by for example Azmat et al. (2013). This constitutes the foundation of our first hypothesis.

This study however finds that the opposite is true, finding strong evidence that the Islamic less conventional yield spread is positive for short to medium term maturities. We apply various techniques to determine the characteristics of the yield spread. We model Islamic and conventional zero yield curves using the Nelson-Siegel model (Nelson & Siegel, 1987), estimated using non-linear least squares as well as an approach inspired by Diebold & Li (2006). We then compare the term structure and establish that the spread is positive for 1-7 year maturities over the studied time period (2004–2014). The spread is a measure of the premium associated with the sukuk market. For longer maturities we can observe a convergence and possibly a crossover.

We can also determine on an issue-specific level that the excess yield of Islamic bonds is persistent across the same maturities as above. Using the zero yield curve for the conventional market, we can estimate the arbitrage-free price of a conventional bond with matching cash flows. We then obtain an estimate of the cost of frictions by differencing the observed yield with the arbitrage-free yield.

The second part of this paper investigates if liquidity has explanatory power on the yield spread. We find that during the time period studied, the spread on zero rates has in general reverted towards a persistent level, however with transitory fluctuations. We argue that these fluctuations can, at least in part, be explained by illiquidity issues on the Islamic market. The transitory effects of illiquidity is studied at the market level, as well as on the issue-specific level. Two different liquidity measures are applied at the issue-specific level: the yield bid-ask spread and the proportion of zero returns. We find that the proportion of zero returns can have some explanatory effect on the yield spread thus finding some support for the second hypothesis.

1.1 Outline of paper

The rest of this paper is organised as follows. Section 2 contains some necessary background on the basics of Islamic finance and Islamic capital markets, in particular the Malaysian market. A review of previous research on sukuk and bond market liquidity is presented in section 3. Thereafter follows a short description of the data used in section 4. The theory and methodology that constitute the basis of this study are presented in section 5. There we outline the procedure for estimating the zero yield curves, how we determine the yield spread and discuss how to interpret the spread. An outline for how the illiquidity component is assessed is also included in this section. Section 6 presents the results and our interpretation thereof, while section 7 gives our most important conclusions. Section 8 presents the limitations of this study and presents suggestions for further research.

2 Background

This section gives an overview of the basics of Islamic capital markets and the Malaysian bond market. It is provided as a benefit to a reader unfamiliar with Islamic finance and Malaysian capital markets. It is based on public information and knowledge provided by the central bank of Malaysia, Bank Negara Malaysia (BNM), on its website. For a more profound review of the basics of Islamic finance and Islamic capital markets we refer to, for example, Iqbal & Mirakhor (2013); Bacha & Mirakhor (2013); Omar et al. (2013) and Krichene (2013).

2.1 Basics of Islamic finance

Islamic finance is based on ethical principles deemed compatible with Shariah (Islamic religious law). There are three key prohibitions that differentiate Islamic and conventional finance. Firstly, the prohibition of *riba*, or usury, forbids the charging of interest according to the conventional interpretation of the Quran. Secondly, the prohibition of *maisir* (betting/gambling) and *gharar* (dubious ambiguity/uncertainty) restrict speculation. Moreover, all companies that promote *haram* (sinful) activities, such as gambling, consumption of alcohol or pork, are excluded from the Islamic financial markets. This limitation imposes an ethical screening process of companies and assets traded in the market.

2.1.1 Islamic bonds

Islamic bonds are called *sukuk*. Formally, a *sukuk* is a document representing the value of an asset. The *sukuk* traded on the Malaysian treasury bond market are in practice a certificate of indebtedness. *Sukuk* can be based on a range of different principles and the criteria for whether a principle is considered Shariah compliant vary across countries. The principles relevant for this study are listed in appendix A. Like conventional bonds, *sukuk* have a fixed maturity, a principal payable at maturity and can bear coupons. Conventional bond valuation models are therefore commonly used to value *sukuk* and the regular yield-price relationship prevails (Bank Negara Malaysia, 2014).

The structure at issuance differs between *sukuk* and conventional bonds. As the charging of interest is considered *haram*, a *sukuk* is asset-based. Issuance of conventional bonds entails the exchange of money against a certificate of indebtedness. The issuer will then pay interest (coupons) over the bond's lifetime, followed by the principal at maturity. For a *sukuk* to be considered Shariah compliant, the certificate of indebtedness must be based in the transfer of an asset. Typically, the ownership of an asset is transferred between the issuer and lender so that the lender makes a (pre-specified) profit on the transaction. The profit is then paid out on a pre-agreed deferred payment basis, constituting the coupons and principal of a *sukuk*. From a cash flow perspective, *sukuk* and

conventional bonds therefore only differ in their issuance structure. On the secondary market, the issuance structure is irrelevant to an investor, and a non-Shariah conscious investor should be indifferent between the two cash flows (Bank Negara Malaysia, 2014).

2.2 The Malaysian bond market

The Islamic financial sector in Malaysia operates in parallel with the conventional sector and has grown steadily since the enactment of the Islamic Banking Act in 1983. The stated aim of the Islamic sector, as outlined in the Financial Sector Masterplan (2001) and the Financial Sector Blueprint (2011), is to contribute to the efficiency and effectiveness of the Malaysian financial sector.

The government of Malaysia is committed to establishing an Islamic financial hub. As the largest sukuk market in the world, it attracts many investors as well as drives the product innovation and financial engineering that characterises the Islamic bond markets (Iqbal & Mirakhor, 2013). The conventional sector still constitutes the majority of the Malaysian market and investors can participate in both markets (Bank Negara Malaysia, 2012).

2.2.1 Bonds traded on the Malaysian Sovereign Debt Market

The government of Malaysia issues both conventional and Shariah compliant bonds and bills. Two types of securities are issued on the conventional market and considered in this paper: *Malaysian Government Securities* (MGS) and *Malaysia Treasury Bills* (MTB). On the Islamic market three types of securities are considered: *Government Investment Issues Profit based* (GII Profit based), *Government Investment Issues Murabaha* (GII Murabaha) and *Malaysian Islamic Treasury Bills* (MITB). Sukuk issued before 2001 are based on the *Quard al-Hasan* principle which did not allow trading on the secondary market and therefore not included in our analysis. The principles and issuance structures of sukuk are detailed in appendix A. Income from sukuk and conventional bonds are treated equally under Malaysian tax legislation. Information on specific issues is publicly available on the official Bank Negara Malaysia (BNM) and BNM Bond Info Hub websites.

- **Malaysian Treasury Bills (MTB)** MTBs are bills traded on the conventional market. They always have short maturities, less than one year. MTBs are always zero coupon bonds. They trade on a discount basis.
- **Malaysian Islamic Treasury Bills (MITB)** MITBs are the Islamic counterpart of an MTB. They are always issued as zero coupon bonds with short maturities. The majority of MITBs are issued with one year to maturity and a minority have 9 months to maturity at issuance. No other maturities are issued. MITBs are based on the *Bai al-Inah* principle and actively traded on the Islamic capital market on a discount basis.

- **Malaysian Government Securities** MGSs are conventional bonds issued with a variety of maturities. They are typically coupon bearing with some exceptions. MGS is the only bond type issued by the government of Malaysia eligible for short selling. Shorting MGS follows restrictive regulations.¹
- **Government Investment Issue: Profit-based (GII)** Profit-based GIIs are Islamic bonds issued with a variety of maturities. All GIIs issued between 2001–2005 are zero-coupon bonds based solely on the *Bai al-Inah* principle. In 2005, a coupon bearing bond based on the *Bai al-Inah* and the *Bai Bithaman-Ajil* principles was introduced. A profit-based GII is asset-based and the transactions involved in the process are illustrated in fig. 1.
- **Government Investment Issue: Murabaha (GII Murabaha)** Murabaha GIIs are bonds with a variety of maturities and typically coupon bearing but occasionally zero coupon. The Murabaha GIIs were introduced to the market in 2013. GIIs based on the *Murabaha* principle are always commodity-linked, typically to crude palm oil. The transaction structure is more complex than the profit-based structure and involves more parties. In addition to the investor and the issuer it includes a *Special Purpose Vehicle* (SPV) and the commodity market. Bank Negara Malaysia is the appointed SPV for all GII issues. Murabaha GIIs are by many scholars considered to be less Shariah-compliant than profit-based GIIs (Azmat et al., 2013).

2.2.2 Interpretation of Shariah law

Islamic finance follows Shariah law and frictions on the Islamic bond market arise from the different interpretations of Quran and Sunnah. To overcome these frictions, BNM established a regulatory body in 1997, the Shariah Advisory Council (SAC). The SAC is the only authorised body to ascertain the Shariah compliance of the Islamic financial sector in Malaysia (Bank Negara Malaysia, 2014).

Many Islamic scholars question the Shariah compliance of several products traded on the Malaysian market. In February 2008, the Accounting and Auditing Organisation for Islamic Financial Institutions (AAOIFF) announced that 85% of sukuk were not Shariah compliant, representing the various levels of strictness across countries (Azmat et al., 2013). According to the stricter interpretations all trading of debt, *Bay al-Dayn*, is considered *haram* (Rosly & Sanusi, 1999), or not compliant with Shariah law, thus excluding all bonds and bills traded on the secondary market. Murabaha contracts are considered less Shariah compliant than profit based sukuk based on the *Bai al-Inah* principle. It could affect the demand of sukuk if some Shariah conscious investors refrain from investing in sukuk with features considered less Shariah compliant, as argued by Azmat et al. (2013).

¹No naked short positions are allowed, only members of the principal dealer network can undertake a short selling position. Retail investors cannot take on short positions and all short selling transactions must be for own positions. Short positions are restricted in time, relative amount and to a list of eligible securities (Bank Negara Malaysia, 2005).

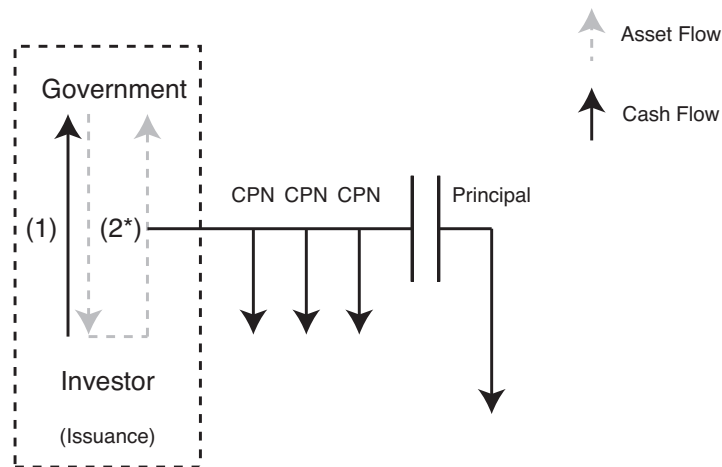


Figure 1: Issuance structure of Bai Bithaman-Ajil sukuk. In the initial transaction (1), the investor purchases an asset from the government for a spot payment. The investor immediately sells the asset back to the government for a (pre-determined) profit (2*) that is paid out in installments (coupons) on a *deferred* basis to the holder of the GII certificate. This certificate can subsequently be traded on the secondary market. The asterisk signifies where the profit transaction takes place. Adapted from Bank Negara Malaysia (2013).

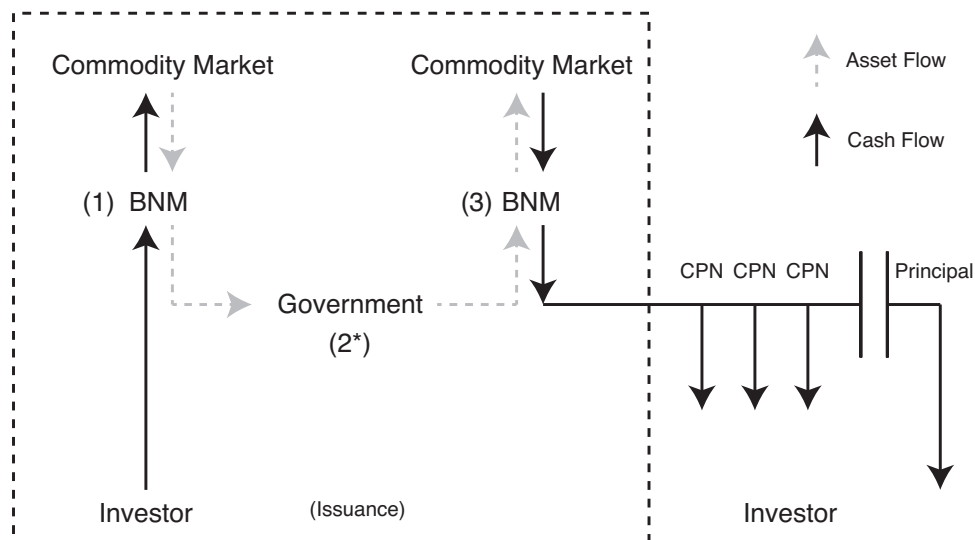


Figure 2: Issuance structure of Murabaha sukuk. In the initial transaction (1), the investor appoints BNM to purchase crude palm oil on the commodity market using funds provided. BNM then sells the commodity to the government on a predetermined cost plus sale basis (2*) that is paid out in installments (coupons) on a *deferred* basis to the certificate holder. The government in turn appoints BNM to immediately sell the commodity on the commodity market (3) in order to raise funds from the transaction. The asterisk signifies where the markup takes place. Adapted from Bank Negara Malaysia (2013).

3 Literature review

The review of previous literature is divided into two sections. Initially it covers an overview of previous research on Islamic finance and capital markets. As published research is limited, a second subsection with a review of relevant research on bond market illiquidity is included.

3.1 Previous research on Islamic Finance

Previous literature on Islamic finance and sukuk markets is relatively limited and often confined to niche publications. Research can generally be divided into two categories; research conducted from a Shariah perspective focusing on the Shariah compliance of securities, and research comparing sukuk and conventional bonds. The latter is often concerned with comparing performance, possible diversification efforts and finding appropriate valuation models for sukuk.

Godlewskic et al. (2013) conduct an event study on the Malaysian market to compare the stock market's reaction to announcements of conventional and Islamic bond issues. The research is thus restricted to the corporate bond market. Fathurahman & Fitriati (2013) compare the yields of sukuk and conventional bonds in Indonesia and find that the yields on sukuk on average are higher than on conventional bonds. Ramasamy et al. (2011) investigate the convexity and duration of sukuk compared to conventional corporate and treasury bonds. The authors conclude that corporate sukuk are less risky than conventional bonds and carry higher risk than treasury bonds.

The scope of this paper is limited to the sovereign debt market, comparing the yield of Islamic and conventional government bonds. A paper written by Safari et al. (2013), and published in a variety of smaller publications, conducts comparisons on sukuk and conventional bonds on the Malaysian bond market. The study includes government and corporate issues, using monthly yield to maturities from August 2001 to April 2012. The research question is similar to the focus of this paper but includes corporate bonds and is not concerned with liquidity. The authors observe a positive yield spread and conclude that sukuk and bonds should be regarded as different instruments.

Azmat et al. (2013) use theoretical modelling to argue that sukuk should carry lower yield. Shariah conscious investors should accept the lower yield and drive up prices to accommodate their ethical preferences. The paper is however confined to theory and never applied to real data.

Another study comparing the performance of sukuk and conventional bonds on the Malaysian market was conducted by Mosaid & Boutti (2014). Comparing series of indices from Bond Pricing Agency Malaysia, the study is not restricted to the treasury market. Observing significant correlation between the return of bonds and sukuk and that sukuk generally outperform conventional bonds, they argue sukuk and bonds are similar in nature. However, they do not attempt to explain the differences in performance.

Previous research, notably Safari et al. (2013), often suggests that sukuk and bonds are different

types of securities. In this paper we argue that once a sukuk has reached the secondary market, a non-Shariah conscious investor is indifferent between the issuance structures and only concerned with the cash flows and risk (in line with views expressed on the BNM Bond Info Hub website).

3.2 Studies on illiquidity and bond yield spreads

Previous literature on sukuk is rarely concerned with liquidity issues. To our knowledge, the effect of illiquidity on the Malaysian sukuk market has never been studied and little research on Malaysian treasury market liquidity has been conducted. The importance of illiquidity in asset pricing is however well recognised in academia. Previous research on liquidity premia and their relationship to the pricing of bonds outside Malaysia is vast and thus the foundation of this paper, most notably Fleming (2003) and Chen et al. (2007).

Fleming (2003) investigated the treasury market liquidity in the US and tested a variety of liquidity measures. The price impact coefficient was found marginally superior to the average bid-ask spread. Three criteria for a good liquidity measure were proposed: simplicity of calculations, availability of data, and high correlation with market participants' view on liquidity as well as ability to proxy transaction costs associated with illiquidity. By these criteria the bid-ask spread showed to be a reasonable tool for tracking and measuring treasury market liquidity.

Looking at the corporate bond market in the US, Chen et al. (2007) investigate how illiquidity is priced in yield spreads by applying two model independent measures: the bid-ask spread and proportions of zero returns, as well as a measure estimated with a model proposed by Lesmond et al. (1999). The authors find that all liquidity measures explain a significant part of the yield spread. Our paper draws heavily on the statistical methods applied for assessing the illiquidity effect.

Bao et al. (2011) investigate US corporate bond market illiquidity and particularly the implication for asset pricing. They investigate the implication of illiquidity on an aggregate as well as issue-specific level. The model applied is constructed to capture the transitory component of illiquidity and its effect on the pricing of an asset. Despite its transitory nature, the liquidity measure can show a degree of persistence depending on the time horizon studied. The authors argue that the bid-ask spread fails to fully gauge the impact of illiquidity on asset pricing.

4 Data

The time period investigated is from January 2004 till March 2014 and we consider all active government bonds in this time period. Our data set covers 832 Malaysian government issues across four classes: MTB, MITB, MGS and GII. The data is summarized in table 1.

Table 1: Aggregates of data collected, by bond type (and Islamic principle). Our data covers all active bonds 2004–2014. *Price observations* is the total number of daily price observations during this period (after filtering). *Coupon* shows the minimum and maximum coupon rates observed over the time period.

Type	Issues	Price Observations	Maturity (Years)	Coupon (%)	Outstanding (MMYR)**
MGS	89	39130	2–30	2.51–9.03	31558
GII	62	12208	2–20	0.00–4.94	28400
<i>Quard al-Hasan*</i>	1	-	3	-	-
<i>Bai al-Inah</i>	5	261	3–7	-	-
<i>Bai Bithaman Ajil</i>	49	11462	3–15	3.08–4.64	25850
<i>Murabaha</i>	7	485	3–20	3.39–4.94	2550
MITB	189	2883	≤ 1	-	200
MTB	492	3805	≤ 1	-	232
Total	832	58026			60390

* Not traded on the secondary market; matured on 2004-06-15.

** Sum of amount outstanding on each issue as reported by Bloomberg, as of 2014-03-14.

The data required to construct zero rate curves consists of issuance and maturity dates, coupon payment dates, coupon rates and prices. Issuance and maturity dates as well as coupon rates are downloaded from Bloomberg. The coupon payment dates are assumed to follow a semi-annual, adjusted for end-of-month schedule. The day count basis for all coupon bearing bonds is actual/actual.

Malaysia's credit rating has not changed during the time period. While a miniscule minority of bonds are rated, all observed ratings are either A (Fitch and S&P) or A3 (Moody's) and do not change over the time period.

Approximately 62 000 daily prices were originally collected from three sources. The primary source is Bloomberg, which accounts for 91.8% of the prices quoted. The Bloomberg dataset includes daily open, close, high, low, bid and ask prices. Unfortunately, Bloomberg data is poor for many MITB and MTB issues, especially before 2009. Therefore, we also use a robot script to download public data from the websites of Bank Negara Malaysia (for early MGS and GII) and BNM Bond Info Hub

(for MITB and MTB). These datasets lack bid- and ask prices.

We drop price observations before the issue date (493 obs.), on or after the maturity date (15), negative yields (75) and yields above 200% (1). All these outliers were clustered to a few dates and issues.

Bonds, especially Islamic issues, are thinly traded. We therefore consider data on a weekly basis. If a closing yield is not available on Friday, we use the average of the bid-ask, if that is available. If not, we instead use the Thursday equivalent. In this way, we may carry forward yields up to four days (Monday). If the most recently available quote is older than four days, we treat the data as missing. This problem is a lot more common in the 2004–2007 time period, and some implications of doing this are discussed in section 8. This results in 12 512 conventional and 5 335 Islamic weekly price observations.

4.1 Market quotes: Discount, Clean Price and Bond Equivalent Yield

All bonds are normalized to a face value of 100 Malaysian ringgit (MYR). MGS and GII are quoted in clean prices, whereas discount bonds (MITB and MTB) are quoted by discount on an actual/360 basis (by Bloomberg) or bond equivalent yield on an actual/actual basis (by BNM Bond Info). In some cases, Bloomberg quotes the price directly. From the market quotes, we compute the semi-annually compounding actual/actual yield as of the settlement date for the closing, bid and ask quotes.

The conversions (as implemented by MATLAB) between discount and bond equivalent yield adhere to the correction for maturities longer than 182 days in (McDonald, 2006, p. 245).

5 Theory and Methodology

The presence of conventional investors in the Islamic market suggests that any cross-market arbitrage opportunities should be eliminated. We therefore construct our hypotheses and method based on typical no-arbitrage arguments. Our method is to construct yield curves from market data collected on sukuk and conventional bonds, respectively. We then compare these over time to determine if there are significant differences between the implied zero rates on the respective markets. Then, we use a no-arbitrage pricing model and the conventional yield curve to derive “arbitrage-free” yields for the Islamic issues, which we can directly compare with market data. Based on these comparisons – between Islamic and conventional zero rates and between Islamic market and “arbitrage-free” yields – we develop two regression models that utilize illiquidity as an explanatory variable for the observed yield spread.

This section begins with developing the two hypotheses that are investigated in this paper. Following this, we explain the basic valuation concepts and notation used in the remainder of the paper. Thereafter, we detail how we construct yield curves using market data. We then outline how we measure the yield spread and how it theoretically is related to market frictions. The last subsection explains the construction of our regressions and liquidity measures.

5.1 Hypothesis development

Hypothesis 1: *The yield of a sukuk should be equal to or lower than the yield of a comparable conventional bond.*

Sukuk differ from conventional bonds in how they are issued and the more complex issuance structure is likely to lead to larger transaction costs for the issuer. However, an investor trading at the secondary market should not be affected by costs associated with issuance. Like conventional bonds, sukuk can bear coupons, have a fixed term maturity and the regular yield-price relationship prevails. Furthermore, the counterparty risk does not vary as all Malaysian government bonds are ranked pari passu (Bank Negara Malaysia, 2014). Therefore, in an arbitrage-free setting, a sukuk and a conventional bond with the same cash flows (and counterparty risk) should have the same fundamental value.

Sukuk should be preferred by Shariah conscious investors who might be willing to accept a lower yield in return for Shariah compliance, thus exerting a downwards pressure on the yield of sukuk. However, unconstrained investors should not value Shariah compliance and therefore be indifferent between conventional and sukuk investments of equal yield. Since shorting of sukuk is prohibited, arbitrageurs would not be able to finance a long position in conventional bonds using a short position in higher yielding sukuk. However, shorting conventional Malaysian bonds (MGS) is possible so that a positive yield spread on Islamic issues should be brought down by arbitrageurs. Therefore, sukuk

should not have consistently higher yields than otherwise equal conventional issues, but may carry lower yields.

Hypothesis 2: *Illiquidity issues can cause a positive yield spread between Islamic and conventional bonds.*

In contradiction to hypothesis 1, previous studies have observed a positive yield spread between sukuk and conventional issues (Safari et al., 2013; Fathurahman & Fitriati, 2013). In our results, we also find strong evidence against hypothesis 1. Previous literature suggests that this spread can be explained by unfamiliarity with the concept of sukuk, interest rate risk, regulatory risks, “Shariah compliance risk”, and/or illiquidity (Krichene, 2013). The presence of arbitrageurs across Islamic and conventional markets means that the spread should be explained by differential frictions. These frictions should be reflected in a liquidity premium, as they essentially relate to an investor’s ability to realize the value of their investment before maturity. Sukuk markets are known to be plagued by illiquidity issues (see, among others, Cakir & Raei (2007) and Jobst (2007)). We therefore argue that illiquidity can have an explanatory effect on the yield spread between sukuk and conventional issues with identical counterparty risk and similar duration.

5.2 Bond valuation, absence of arbitrage and the yield curve

We analyze discount (MITB and MTB) and semi-annual fixed coupon bonds (GII and MGS) in this paper. Both sukuk and conventional bonds are fixed-interest securities with a definite maturity and payment schedule. Thus, we apply the same basic valuation models to both types of securities. Absence of arbitrage requires that the price of a coupon-bearing bond be equal to the price of a portfolio of zeros reconstructing the cash flow of the coupon bond (Veronesi, 2011). At time t , we discount all future cash flows using a τ -specific discount rate $Z_t(\tau)$:

$$P_t = \sum_{\tau=0.5}^T Z_t(\tau)C + Z_t(T)M \quad (5.1)$$

Where C is the semiannual coupon payment and M is the principal redeemed at maturity.

It follows from equation (5.1) that by finding Z_t , we can determine the arbitrage-free price of any coupon bond. Using continuously compounding rates, the functional form of Z_t is:

$$Z_t(\tau) = \exp(-r_t(\tau)\tau) \quad (5.2)$$

Where $r_t(\tau)$ is the continuously compounding yield on a τ -year zero coupon bond at time t . The *yield curve* formed by $r_t(\tau)$ represents the term structure of the economy at time t . A complete

description of the zero yield curve can therefore be used with eq. (5.1) to price any coupon bearing bond. Typically upwards sloping, the yield curve reflects both expectations of future interest rates and the “term premium” demanded by investors for investing in long rather than short term assets (Kim & Orphanides, 2007).

Two cash flows of equal risk and maturity T should have equal price. Equivalently, their yield to maturity should be equal. Under hypothesis 1, the Islamic yield curve should therefore be equal to, or below, the conventional equivalent.

5.3 Modelling the yield curve

Beyond short maturities, neither $Z_t(\tau)$ nor $r_t(\tau)$ are directly observable as Malaysia generally only issues zero coupon bonds with maturities up to one year. The construction of the yield curve beyond 1 year maturities therefore requires estimation and modelling based on market data. In this paper, we use the functional form of the yield curve introduced by Nelson & Siegel (1987), estimated using both a non-linear least squares technique and a “hybrid” model based on bootstrapping and Diebold & Li (2006).

5.3.1 Nelson-Siegel model

Introduced by Nelson & Siegel (1987), the Nelson-Siegel model is commonly used for modelling the yield curve. The model generates smooth curves and can take on the shapes that are commonly associated with yield curves. The model fits the yield curve using the functional form:

$$r_t(\tau) = \beta_0 + \beta_1 \left(\frac{1 - e^{-\lambda\tau}}{\lambda\tau} \right) + \beta_2 \left(\frac{1 - e^{-\lambda\tau}}{\lambda\tau} - e^{-\lambda\tau} \right) \quad (5.3)$$

Where $r_t(\tau)$ is the continuously compounding zero yield with τ years to maturity. β_0 , β_1 , β_2 and λ are the estimated model parameters. The respective factors associated with each β parameter are referred to as “factor loadings”. Nelson & Siegel (1987) introduced an intuitive interpretation of the model parameters. The constant, β_0 , governs the long term yield level while the β_1 factor governs the short-term yield as it starts at one and thereafter monotonically decreases. β_2 controls the medium term yield as it reaches its maximum at the medium term. λ governs the decay rate – how quickly β_1 and β_2 “drop off” with increasing τ . As discussed by Diebold & Li (2006), the parameters β_0 , β_1 and β_2 can also be interpreted as representing level, slope and curvature of the yield curve.

A popular extension to the Nelson-Siegel model was introduced by Svensson (1995) using two additional parameters (β_3 and λ_2) in order to better model highly non-linear yield curves. While this improves the model’s ability to fit data, the cost of two additional parameters means that the

model cannot be reliably estimated in situations with poor data – a situation which unfortunately characterizes the Malaysian bond market quite well.

5.3.2 Non-linear least squares estimation of Nelson-Siegel parameters

A non-linear least squares estimation procedure attempts to minimise the objective function

$$\Omega(\beta_0, \beta_1, \beta_2, \lambda) = \sum_{i=1}^n (\text{Actual price}_i - \text{Model price}_i)^2 \quad (5.4)$$

where Model price_i is the arbitrage-free price of bond i as determined by using eqs. (5.1) to (5.3) and Actual price_i is the observed price of the bond. Technically, the β and λ parameters vary over time and are therefore estimated for each week; we drop the time subscripts here for convenience.

We refer to curves estimated in this manner as “Nelson-Siegel” curves throughout this paper. The estimation procedure is performed using the `IRFunctionCurve.fitNelsonSiegel` procedure in MATLAB.

5.3.3 Bootstrapping and the Diebold-Li model

Bootstrapping of yield curves is an iterative process to obtain zero rates using market data and already calculated zero yields. From a zero coupon bond actual price we can directly obtain the prevailing zero rate:

$$Z_t(\tau) = \frac{M}{\text{Actual price}_t} \quad (5.5)$$

We can derive from equation (5.1) and (5.5) that the zero discount factor from time t to time T can be estimated from a coupon bearing bond:

$$Z_t(T) = \frac{\text{Actual price}_t - \sum_{\tau}^{T-0.5} Z_t(\tau)C}{M_T + C} \quad (5.6)$$

Curves are bootstrapped using an iterative solution algorithm implemented in `IRDataCurve.bootstrap` in MATLAB. This process insists that all bonds used for the estimation are perfectly priced by the yield curve while non-observable yields are calculated using simple linear interpolation (Hagan & West, 2008).

Bootstrapping does not guarantee that yield curves are “smooth” or are useful for extrapolation. Therefore, we never use bootstrapped curves directly, but instead as inputs for estimating the parameters of the Nelson-Siegel model using OLS as inspired by Diebold & Li (2006).

By fixing λ at a specific value, Diebold & Li (2006) find that the “factor loadings” on each β parameter is dependent only on the time to maturity τ . Given a sample of zero rates, the remaining

three parameters of eq. (5.3) can be estimated using OLS. In their original article, the authors use “unsmoothed Fama-Bliss forward rates” as the basis for their estimation. These are unfortunately unavailable for the Malaysian market.

We therefore regress zero rates obtained by bootstrapping onto the Nelson-Siegel model using OLS. In practice, this generates β estimates that are typically less extreme and less varying over time than the non-linear least squares method – making them more “meaningful” when interpreted as level, curvature and slope. The “hybrid” curves are analogous to smoothed versions of the bootstrapped curve. The bootstrapped curves are thus redundant for our analysis apart from their application in estimating the hybrid curves. We refer to the “hybrid” curves as Diebold-Li curves for the remainder of this paper.

Diebold & Li (2006) propose fixing λ so that the loading on the β_2 parameter, the medium term factor, is maximized at 30 months. Our unit of τ is, however, years and we correspondingly fix λ so as to maximize the loading at $\tau = 2.5$ giving $\lambda = 0.7573$.

5.4 Determining the yield spread

5.4.1 The yield curve spread

The yield spread is simply the spread between the Islamic and conventional zero rate. This spread represents the Islamic-conventional yield spread at the market level. The construction of the yield curves returns an estimate of the prevailing zero rate at time t for any maturity T .

$$\text{Yield Spread}_{t,T} = r_{\text{Islamic},t}(T) - r_{\text{conventional},t}(T) \quad (5.7)$$

Where $r_{\text{Islamic},t}(T)$ is the T -year zero rate on the Islamic market, and $r_{\text{conventional},t}(T)$ its conventional counterpart. As argued in hypothesis 1, the yield on a sukuk should be equal to or lower than the yield of a comparable conventional bond. We can test this hypothesis by doing a one-sided paired t test on the yield spread.

5.4.2 Issue specific yield spread

When considering yield curves at the market level all issue-specific characteristics are lost. Zero rates are also effectively “fictional” as non-coupon bonds beyond 1 year are generally unavailable in the Malaysian bond market. Moreover, since sukuk are thinly traded, the estimation of yield curves based on market data can easily result in unreliable estimates. In contrast, data on conventional bonds is highly available. Ideally, we would like to compare pairs of otherwise identical Islamic and conventional bonds. However, as few if any bonds match each other exactly in this regard, we propose using comparisons based on arbitrage-free yields derived from the conventional yield curve.

Consider the yield curve generated from the conventional dataset. At any point in time we have an estimated zero rate for each maturity. By discounting the promised cash flows of a sukuk using the conventional curve and eq. (5.1), we can derive the arbitrage-free price of a conventional bond with identical cash flows:

$$\text{Arbitrage-free price}_{it} = \sum_{\tau=0.5}^T Z_{\text{conventional},t}(\tau)C + Z_{\text{conventional},t}(T)M \quad (5.8)$$

$Z_{\text{conventional},t}(\tau)$ is the zero discount factor from time t to time τ obtained from the estimated conventional yield curve. This price should, in principle, represent the fair price of a conventional bond with cash flows identical to the sukuk, assuming no-arbitrage and a perfect model of the yield curve.

The arbitrage-free price calculated using eq. (5.8) should therefore represent the price of a sukuk absent frictions, relative to a conventional bond. Differencing the observed price of a sukuk with the arbitrage-free price generates the magnitude of the price frictions associated with the sukuk market. The price friction can conveniently be expressed in a yield spread:

$$\text{Yield Spread}_{i,t} = \text{Observed Yield}_{i,t} - \text{Model Yield}_{i,t} \quad (5.9)$$

Where $\text{Observed Yield}_{i,t}$ is the yield of sukuk i obtained from our dataset and $\text{Model Yield}_{i,t}$ is the yield derived from the arbitrage-free price.

Note that the Nelson-Siegel model is *not* a perfect model of the yield curve. Therefore, the no-arbitrage condition will be violated even when pricing conventional issues with the conventional yield curve. Coroneo et al. (2008) test the “statistical” no-arbitrage properties of the Nelson-Siegel model and find that it performs well in this regard. Analogously, as a form of model validation, we first statistically test how well our estimated yield curves fit their own data. Then, we perform the same tests of the deviations of Islamic market yields from the arbitrage-free yields of a conventional bond with matching cash flows to determine whether the yield spread is statistically significant.

5.5 Liquidity and frictions

Our second hypothesis proposes that an observed yield spread should be explainable by illiquidity, or frictional, factors in the Islamic market. The remainder of this subsection is concerned with specifying these factors and the regression models used. The applicable liquidity measures are specified in the final subsection.

Illiquidity should by its nature be transitory, as argued by Bao et al. (2011). However, over a limited time horizon, as in the case of this study, it is feasible to observe a persistent level of frictions.

If a shock “pushes” the two interest rates apart, the presence of arbitrageurs should ensure that the spread reverts to its average level as determined by the cross-market frictions (Wooldridge, 2009, see p. 615). Basic economic intuition therefore requires that we can discard the existence of a unit root in the yield spread time series. By performing a Dickey-Fuller test we can check this basic economic requirement.

5.5.1 Market Liquidity

In principle, using ordinary least squares, we model the spread between Islamic and conventional zero rates as a simple linear function of frictions:

$$\text{Yield Spread}_t = \text{Islamic Yield}_t - \text{Conventional Yield}_t = \beta_0 + \beta_1 \text{Liquidity}_t + \varepsilon_t \quad (5.10)$$

Interpreting the yield deviations as a function of frictions gives eq. (5.10) an intuitive interpretation: β_0 is the persistent element of frictions, while β_1 is the transitory liquidity element.²

Zero rates at time t are highly likely to be correlated with the zero rates in the previous time period. Serial correlation in the error term might therefore lead to biased estimates of eq. (5.10) (Wooldridge, 2009; Tsay, 2005). This issue should be largely circumvented by estimating the first differenced equation instead:

$$\Delta \text{Yield Spread}_{t,t-1} = \beta_1 \Delta \text{Liquidity}_{t,t-1} + u_t \quad (5.11)$$

The β_1 parameter should now capture the transitory liquidity component while the persistent element is differenced away.

This model is however likely to suffer from severe omitted variable bias. Particularly difficult for its economic interpretation is the possible presence of negatively confounding variables omitted from the equation. At the market level, we find it impossible to measure such factors with the available data.

In our results, we estimate eq. (5.11) for six maturities on a weekly basis: The 1, 2, 3, 5, 7 and 10 year zero rates as obtained by the yield curves. The regressions are run week-to-week over the time period where we have sufficiently reliable data.

²A more formal treatment of the causal relationship between the conventional and the Islamic zero rates would be to model both as endogenous variables in a vector autoregression (cf. Tsay (2005)), however, doing so falls outside the scope of this paper.

5.5.2 Issue-specific Liquidity

Consider the price frictions obtained from our pricing model in eqs. (5.8) and (5.9). Remember that $\text{Yield Spread}_{i,t}$ represents the spread between the observed yield of a sukuk and the arbitrage-free yield on a conventional bond with matching cash flows. We estimate the following (pooled) cross-sectional regression which is largely inspired by Chen et al. (2007).

$$\begin{aligned} \text{Yield Spread}_{i,t} = & \beta_0 + \beta_1 \text{Liquidity}_{i,t} + \beta_2 \text{Maturity}_{i,t} + \beta_2 \text{Coupon}_i \\ & + \beta_3 \text{Slope}_t + \varepsilon_{i,t} \end{aligned} \quad (5.12)$$

Where $\text{Liquidity}_{i,t}$ is a measure of liquidity (specified in section 5.6), $\text{Maturity}_{i,t}$ is the bond's time to maturity (in years) and Coupon_i is its coupon rate (in percentages). Slope_t is calculated as the 10 year less the 2 year interest rate of the conventional yield curve. Chen et al. (2007) also control for some other factor such as equity volatility and issuer-specific characteristics which are not applicable to the government issues used in this paper.

We estimate multiple versions of eq. (5.12): As a univariate regression solely on the liquidity measure; with controls for the other determinants; and with annual time dummies.

The choice of annual time dummies is a practical compromise. The purpose of time dummies is to control for changes in the distribution between samples over time (Wooldridge, 2009). Besides the econometrical issues with estimating a model with 500 parameters, there is little reason to believe that such changes would be significant from one week to another. The choice of annual time dummies is admittedly also somewhat arbitrary (and coarse), but should allow us to control for some bias.³

Pooled OLS is preferred over a fixed effects or first difference estimation due to the presence of serial correlation and the extremely unbalanced panel.

5.6 Choice of liquidity measure

The choice of liquidity measure is dependent on whether we look at market level or on specific issues. At the market level we need a proxy of the prevailing aggregate liquidity. Looking at specific issues, individual measurements of liquidity can be considered. However, from both a theoretical and practical perspective, finding a good measure of liquidity is not straightforward.

In all cases, our measures are more precisely described as “illiquidity” measures. The expected effect is that these measures have a widening effect on the spread.

At the aggregate level we apply the average bid-ask spread. Many argue for the inadequacy of the spread as a measure for liquidity (Bao et al., 2011; Chen et al., 2007). Fleming (2003), however,

³In our results, we find that using quarterly dummies produces more reliable estimates; however, using even more frequent controls does not appear to improve the results markedly.

argues that the simplicity of calculations and availability of data still make it a useful proxy for tracking and measuring liquidity. The average (yield) bid-ask spread in a given time period for a given market is given as:

$$\text{Bid-Ask Spread}_t = \frac{1}{n} \sum_{i=1}^n \text{Bid Yield}_{i,t} - \text{Ask Yield}_{i,t} \quad (5.13)$$

For our purposes, we find the average bid-ask spread each week by averaging the end-of-week bid-ask quotes. Note that because the ask *price* is practically always higher than the bid *price*, the reverse is true for yields. We compute the average bid-ask spread both for Islamic and conventional issues separately, as well as an aggregate measure across markets. Due to some obvious data errors, we winsorize these values at the 1% and 99% levels.

Chen et al. (2007) find that issue-specific liquidity measures have a significant explanatory effect on corporate yield spreads. They measure liquidity using three different measures: The bid-ask spread, the proportion of zero returns and a third measure introduced by Lesmond et al. (1999). We restrict ourselves to using the first two.

The bid-ask spread is simply the bid less the ask yield of bond i at time t :

$$\text{Bid-Ask Spread}_{i,t} = \text{Bid Yield}_{i,t} - \text{Ask Yield}_{i,t} \quad (5.14)$$

The proportion of zero returns in a given period is the number of days without any price movements, over the total number of trading days. Lesmond et al. (1999) consider days without trading (no recorded price) and days with no change in price as zero returns. We follow this approach. For thinly traded assets, the zero return measure becomes problematic. Chen et al. (2007) consider the measure “inestimable” if more than 85% of daily returns are zero over a year. We measure the incidence of zero returns per week, counting the number of days the price has changed since the last Friday (not counting weekends).

The link between liquidity and zero returns is typically made through transaction costs (Chen et al., 2007; Lesmond et al., 1999). In essence, zero returns are theorized to occur when the value of new information is less than the cost of trading. In the context of this paper, we use weekly data. Therefore, the proportion of zero returns reflects an uncertainty in the current market price. The weekly market quote of an Islamic issue with a high incidence of zero returns would therefore be more likely to have “out-of-date” (stale) prices.

6 Results

Our results are organized in four subsections. The respective figures and tables are collected at the end of each section.

First, we present the estimation, descriptives and difference testing of zero rate curves across conventional and Islamic markets. Here, we find evidence of a small, but generally positive and stable, yield spread for Islamic over conventional maturities up to about 7 years.

Next, we discuss the fit of these curves to the market data, which serves both as a robustness check of the results and as a basis for the arbitrage-free yield estimation. We show that, in general, the yield curves accurately price their respective markets with a minor bias towards overpricing.

Third, we present the results of estimating the implied arbitrage-free yields on the Islamic issues from the conventional yield curves. We present descriptives for different maturities and types of sukuk. Again, we run difference tests that verify the statistical significance of the mispricing, suggesting a positive yield premium for Islamic issues with less than 7 years to maturity.

The fourth section presents the results of our liquidity estimations and regressions. We find no evidence of liquidity effects at the aggregate level, but the cross-sectional analysis suggests that the incidence of zero returns has a statistically significant contribution to the yield spread.

6.1 Yield curves

The results of estimating yield curves are best seen graphically. The time series of yield curves forms a surface over time (X), maturities (Y) and yields (Z). These surfaces are shown in fig. 4, with one surface for each market and methodology. The most striking feature of these figures are the issues with the Islamic market. Weeks with missing data, when number of quoted trades are less than 5, show up as gaps (especially evident before 2009). Most notable are the extreme short-term outliers even in cases where data is available. These do not reflect measurement errors but rather problems with the estimation procedures in the face of thin data. The conventional curves are in stark contrast to the Islamic ones. Notably, they are very similar in appearance and almost always upwards sloping.

Overlooking the extreme outliers in the Islamic market, we note that, in general, the two markets do seem to follow each other closely. The close correlation between Islamic and conventional markets is made evident by comparing the yields for identical maturities for each week, as shown in fig. 5. Regardless of maturity, the curves generally follow each other tightly.

Figure 5 suggests a number of interesting patterns. In general, it appears that Islamic yields are above conventional yields of equal maturity, for maturities up to 7 years. While the spread fluctuates it appears to return to a persistent level over time. The common level of interest rates is also of note. After having dropped sharply in the beginning of 2009, they have remained relatively stable until

mid-2013 when both Islamic and conventional rates started rising. However, short-term Islamic rates remained more stable than both conventional yields and long-term Islamic rates.

The lack of data and resulting unreliable estimates for the Islamic market pose significant issues for continued analysis. We consider a week as “lacking” data if it does not contain at least five observations, or if the span between the longest and shortest maturity is less than five years. This criterion is motivated by two considerations. First, the Diebold-Li OLS model estimates three parameters, whereas Nelson-Siegel estimates four. Consequently, even five observations is questionable from an econometrical perspective. Second, if observations are too clustered around a single maturity, there is effectively too little variation in the x values for OLS to produce usable estimates and likewise, non-linear least squares will adopt many different local minima.

This restriction excludes nearly all Islamic curves in the period before 2007 (cf. fig. 5). However, data is generally available since mid-2007 and forwards, with almost continuous data available since 2009. Accordingly, the remainder of the results in this subsection only considers the seven year period 2007–2014 which, while admittedly short, should cover approximately one business cycle.

We can now present representative descriptives of the Islamic and conventional yields in table 2. The close correlation between markets is shown in table 3. The extremely high correlations between both parameters and interest rates supports a hypothesis that Islamic and conventional yields are driven by similar underlying factors.

The long-term β_0 parameter, representing the level of the yield curve, is lower for Islamic issues over the time period and the yield spread is generally positive up until 7 year maturities. Also, it appears that the positive yield spread for shorter maturities is largely attributable to different curvatures in the yield curve as captured by β_2 (which, not surprisingly, is the parameter with the smallest cross-market correlation).

Nelson-Siegel descriptives are also shown in table 2. While the estimated yields are generally close to Diebold-Li estimates, the parameters of the model fluctuate wildly. Having found that in general, the Nelson-Siegel and Diebold-Li models produce similar fits to the data, we proceed mainly with presenting results based on the Diebold-Li model. Besides typically producing more realistic short term estimates with little data, it has the attractive property of having more consistent, and therefore useful, β estimate values. Nelson-Siegel results can be found in appendix B. We also perform more careful model validation in section 6.2.

The “average yield” curves in fig. 3 is another illustration of how the markets relate to each other (2007–2014). We see that the average Islamic yield is higher up to 7 years maturity, after which it reverses. This is consistent with the descriptives in Table 2.

6.1.1 Statistical yield curve tests

As outlined in section 5.5.1 we check the respective time series of interest rates and spread for unit roots (see table 4). As is commonly seen in interest rate time series, we cannot disprove a unit root for most maturities of both conventional and Islamic zero rates. Conversely, the same hypothesis is strongly rejected for the spread of all maturities, which is necessary for the long-run absence of arbitrage.

We test statistically the significance of the differences between markets. The results are shown in table 2. These tests suggest that the differences are highly significant. At virtually any significance, both the one- and two-sided (that Islamic zero rates are less than or equal to conventional equivalents) hypotheses are rejected.

These results are very interesting. They suggest a statistically significant yield premium from investing in Islamic bonds over conventional bonds, which goes against the first hypothesis of this paper. At least for short to medium term investments, it appears that Islamic investors are compensated for their choice of Shariah compliant investments. The differences seem to cluster around 5-10 basis points – a difference which should be considered also economically significant for zero rates of equivalent maturity and counter party risk. These findings, both in magnitude and direction, generally agree with the findings of Safari et al. (2013).

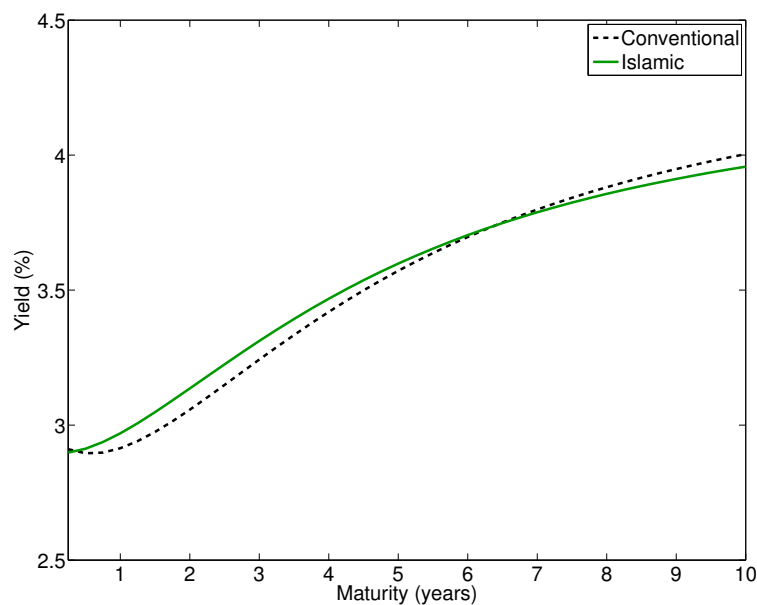
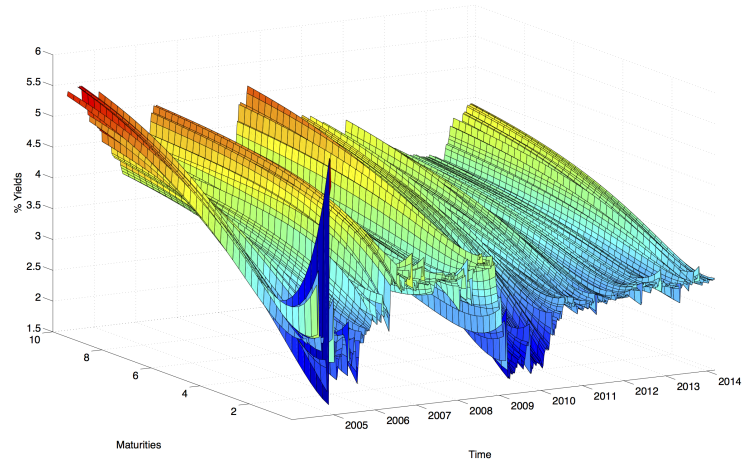
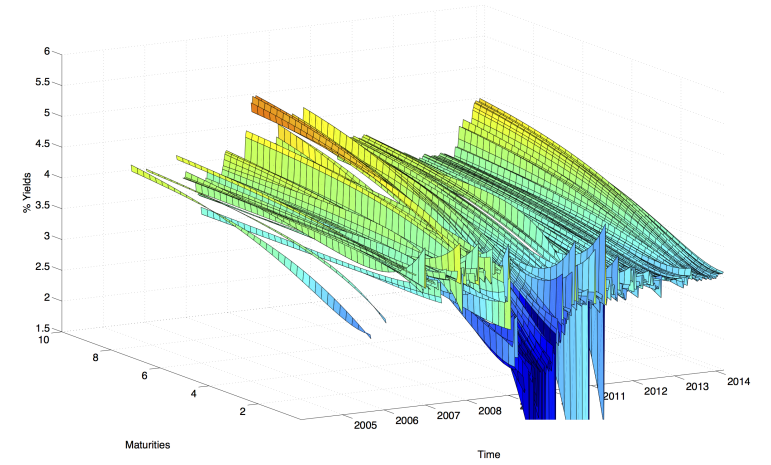


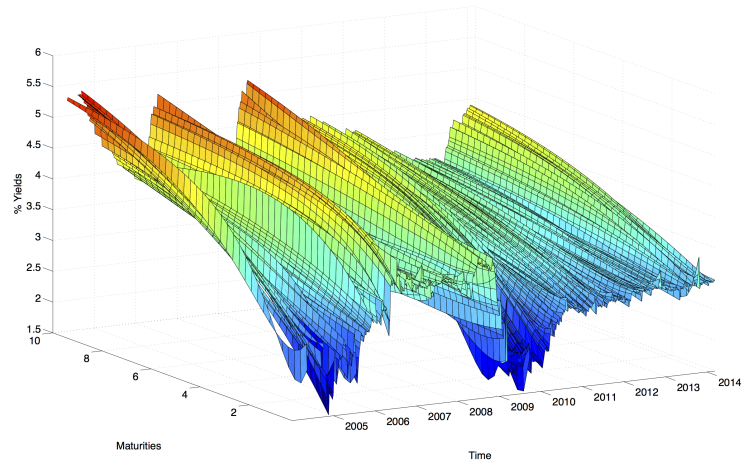
Figure 3: Yield curves formed by taking the time averages of Diebold-Li β estimates for Islamic (solid green) and conventional (dashed black) markets (2007–2014). Excludes weeks with less than five observations or too few maturities.



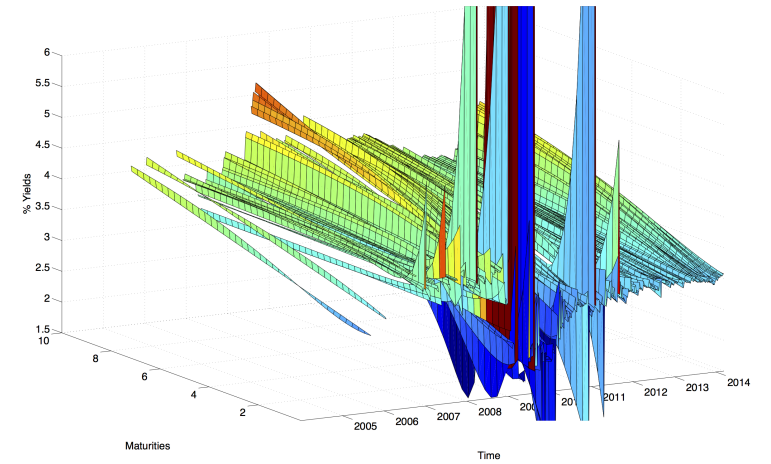
(a) Conventional, Diebold-Li



(b) Islamic, Diebold-Li

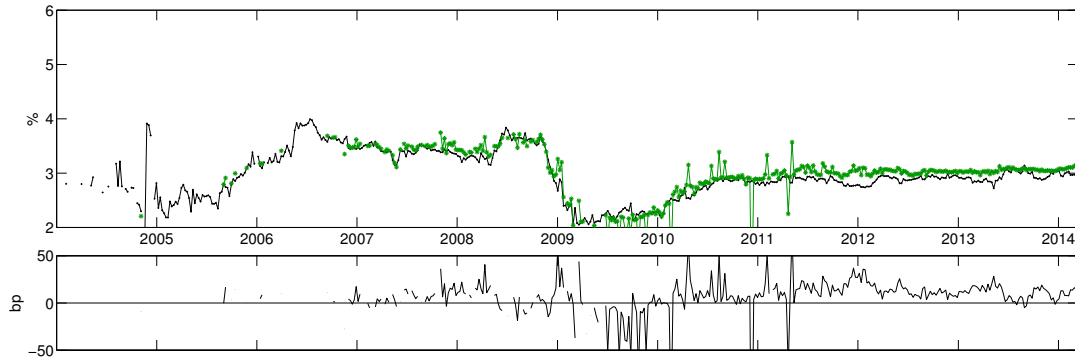


(c) Conventional, Nelson-Siegel

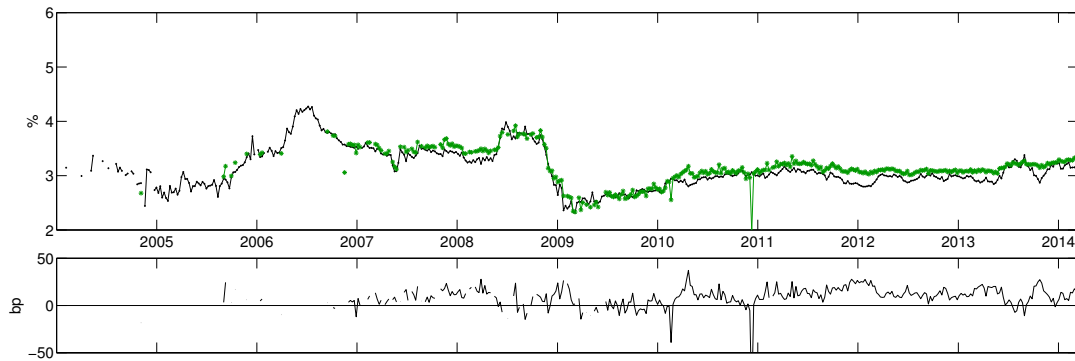


(d) Islamic, Nelson-Siegel

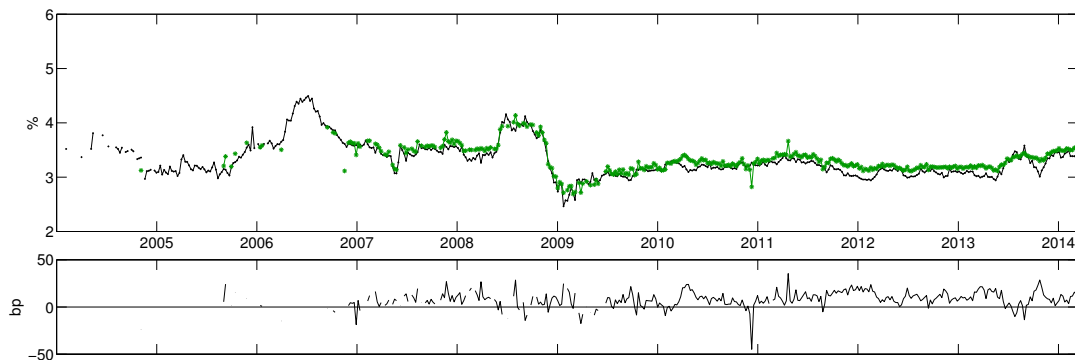
Figure 4: Yield curve surfaces showing time 2004-2014 (X), 0-10 year maturities (Y) and yield (Z). Only shows weeks with at least five observations and minimum five years between shortest and longest maturities but without any other filtering.



(a) 1 year

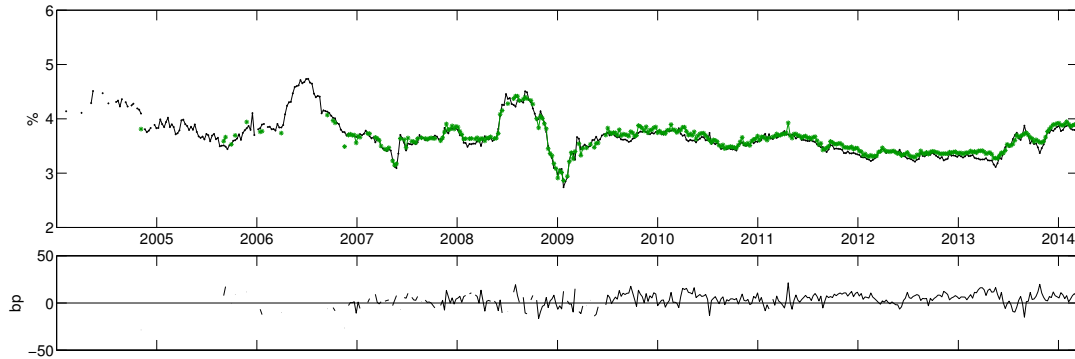


(b) 2 year

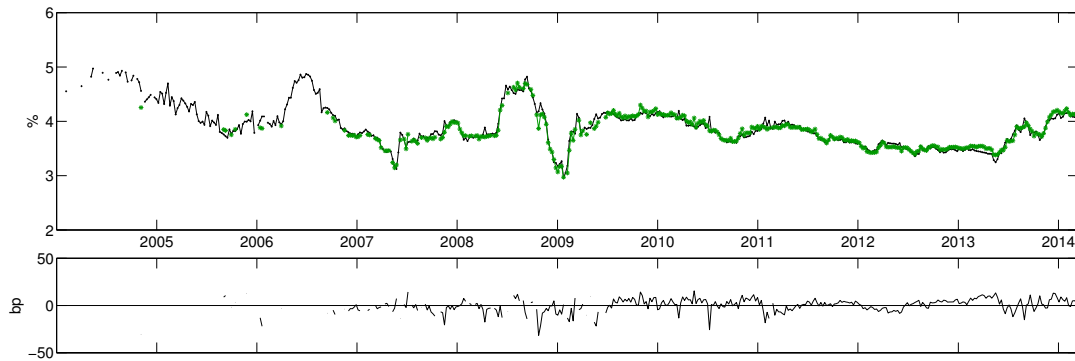


(c) 3 year

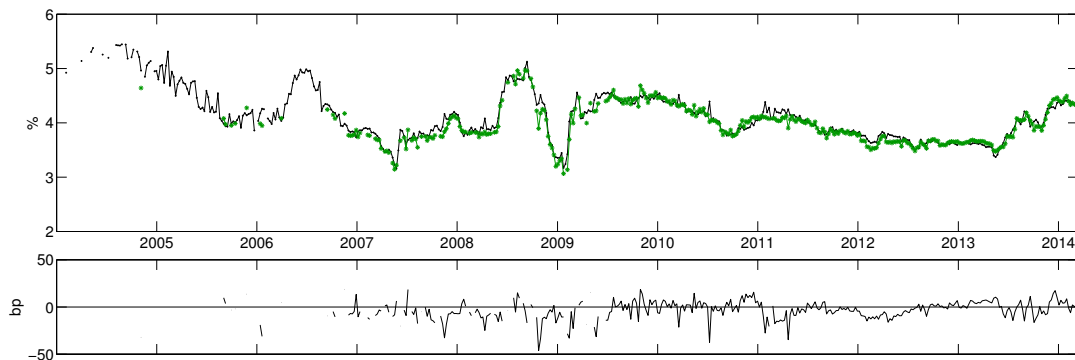
Figure 5: Time series of Diebold-Li yield curves at selected maturities. The upper panel shows Islamic (green) and conventional (black) on top of each other. The bottom panel shows the spread, Islamic less conventional, in basis points (bp). Only shows weeks with at least five observations and minimum five years between shortest and longest maturities (missing weeks appear as gaps). *Continued on next page.*



(d) 5 year



(e) 7 year



(f) 10 year

Figure 5: *Continued.* Time series of Diebold-Li yield curves at selected maturities. The upper panel shows Islamic (green) and conventional (black) on top of each other. The bottom panel shows the spread, Islamic less conventional, in basis points (bp). Only shows weeks with at least five observations and minimum five years between shortest and longest maturities (missing weeks appear as gaps).

Table 2: Yield curve descriptives and difference tests. 2007–2014 after excluding weeks with less than five observations or less than five years between longest maturities. There are $N_c = 367$ conventional weeks, and $N_i = 339$ Islamic weeks. The one-sided t-test null hypothesis is that the Islamic value is less than or equal to the conventional one.

	Islamic				Conventional				Difference		T-test p-values	
	Mean	S.D.	Min	Max	Mean	S.D.	Min	Max	Mean	S.D.	1-sided	2-sided
Diebold-Li												
β_0	4.380	0.540	3.144	5.656	4.520	0.528	3.244	5.893	-0.1277	0.0105	1.000	0.000***
β_1	-1.483	1.108	-6.919	0.636	-1.571	0.902	-3.729	0.099	0.0909	0.0268	0.000***	0.001**
β_2	-1.560	1.050	-5.529	4.736	-2.147	0.802	-6.862	0.258	0.5905	0.0557	0.000***	0.000***
1YR (%)	2.970	0.447	0.459	3.745	2.914	0.391	2.052	3.837	0.0705	0.0118	0.000***	0.000***
2YR (%)	3.136	0.297	1.982	3.920	3.057	0.305	2.315	3.984	0.0935	0.0060	0.000***	0.000***
3YR (%)	3.312	0.233	2.688	4.137	3.243	0.261	2.464	4.158	0.0836	0.0045	0.000***	0.000***
5YR (%)	3.599	0.240	2.864	4.418	3.571	0.264	2.741	4.506	0.0408	0.0034	0.000***	0.000***
7YR (%)	3.789	0.292	2.970	4.705	3.798	0.305	2.961	4.820	0.0033	0.0037	0.185	0.370
10YR (%)	3.958	0.354	3.065	4.969	4.004	0.359	3.156	5.124	-0.0331	0.0051	1.000	0.000***
Nelson-Siegel												
β_0	33.465	80.736	0.001	742.805	22.740	63.615	0.001	465.099	12.2445	5.0811	0.008**	0.017*
β_1	-23.403	135.711	-739.378	1940.278	-12.335	157.503	-462.249	2741.149	-13.1890	11.2018	0.880	0.240
β_2	-20.701	124.549	-1977.205	654.948	-15.214	147.726	-2748.838	175.708	-6.3108	10.8098	0.720	0.560
1YR (%)	3.165	2.571	0.848	48.447	2.931	0.443	1.644	3.901	0.2491	0.1408	0.039*	0.078
2YR (%)	3.140	0.325	1.450	3.911	3.091	0.330	2.210	4.030	0.0638	0.0067	0.000***	0.000***
3YR (%)	3.311	0.244	1.752	4.120	3.260	0.257	2.525	4.153	0.0651	0.0058	0.000***	0.000***
5YR (%)	3.580	0.247	2.852	4.451	3.541	0.269	2.637	4.406	0.0518	0.0039	0.000***	0.000***
7YR (%)	3.766	0.296	2.967	4.708	3.748	0.321	2.799	4.784	0.0318	0.0039	0.000***	0.000***
10YR (%)	3.966	0.351	3.073	5.192	3.966	0.372	3.061	5.231	0.0134	0.0056	0.009**	0.017*

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 3: Correlation between selected conventional (rows) and Islamic (columns) zero rates and yield curve parameter estimates (2007–2014, Diebold-Li model). Excludes weeks with less than five observations or too few maturities.

Islamic	1YR	2YR	3YR	5YR	7YR	10YR
Conventional						
1YR	0.88	0.93	0.80	0.26	-0.04	-0.21
2YR	0.78	0.93	0.92	0.52	0.25	0.07
3YR	0.58	0.81	0.94	0.78	0.56	0.41
5YR	0.11	0.40	0.73	0.97	0.92	0.84
7YR	-0.14	0.14	0.52	0.93	0.97	0.95
10YR	-0.28	-0.02	0.36	0.86	0.96	0.96
Islamic	β_0	β_1			β_2	
Conventional						
β_0	0.93	-0.79			-0.38	
β_1	-0.86	0.90			0.39	
β_2	-0.55	0.51			0.41	

Table 4: Dickey-Fuller tests for unit roots in interest rates and the Islamic less conventional spread (2009–2014). P-value refers to the MacKinnon approximate p-value.

Islamic	1YR	2YR	3YR	5YR	7YR	10YR
Z-statistic	-6.036	-4.446	-4.196	-2.191	-1.510	-1.915
p-value	0.000	0.000	0.007	0.210	0.529	0.325
Conventional	1YR	2YR	3YR	5YR	7YR	10YR
Z-statistic	-1.816	-3.500	-4.429	-2.528	-1.855	-1.851
p-value	0.372	0.008	0.000	0.109	0.354	0.356
Spread	1YR	2YR	3YR	5YR	7YR	10YR
Z-statistic	-12.614	-8.920	-7.855	-8.764	-9.114	-10.029
p-value	0.000	0.000	0.000	0.000	0.000	0.000

6.2 Model validation using arbitrage-free yields

A good model of the yield curve should be able to accurately price the bonds used to estimate it using eq. (5.1). The performance of the models in this regard is evaluated in table 5 and illustrated in fig. 6. We compare the market yield with the arbitrage-free yield derived from each model's yield curve (as outlined in section 5.4.2). This table and figure also present the results of our "cross-market" methodology, the results of which are discussed in the next subsection.

A bond is called underpriced if its market yield is greater than the arbitrage-free yield derived from the relevant yield curve. If the models are accurate, the no arbitrage condition stipulates that this "excess yield" equal zero.

The fit of the models to the data differs on a week to week basis, making it hard to give a measure of the "overall" fit. While the mean excess yield is extremely small (less than -0.5 basis points) for all curves, there is a considerable amount of variance. On a week-to-week basis, we can only in singular cases reject the null hypothesis that the mean excess yield is non-zero (even at typically insignificant confidence levels). All in all, it seems that both Diebold-Li and Nelson-Siegel fit the data quite well.

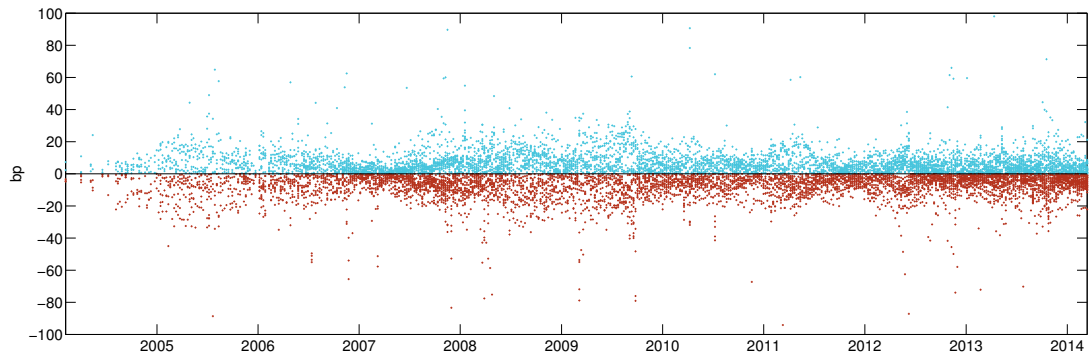
Perhaps most important for the continued analysis in this paper is that the model deviations are "unbiased", so that the occurrence of under- and overpricing is roughly equal. We therefore count the occurrences of under- and overpricing and find that the models perform well also in this regard. The proportion of under- and overpricing is roughly 50/50, with a minor bias in the direction of overpricing over time. The symmetric distribution around zero for each model is especially evident by looking at fig. 6.

One could evaluate many additional criteria for the model fit, for example setting a "tolerance level" that no observed yield must differ by more than x basis points from the arbitrage-free yield. As with the zero rate curves, the arbitrage-free models also suffer from outliers. Incidences of outliers are however occasional and not biased in either direction.

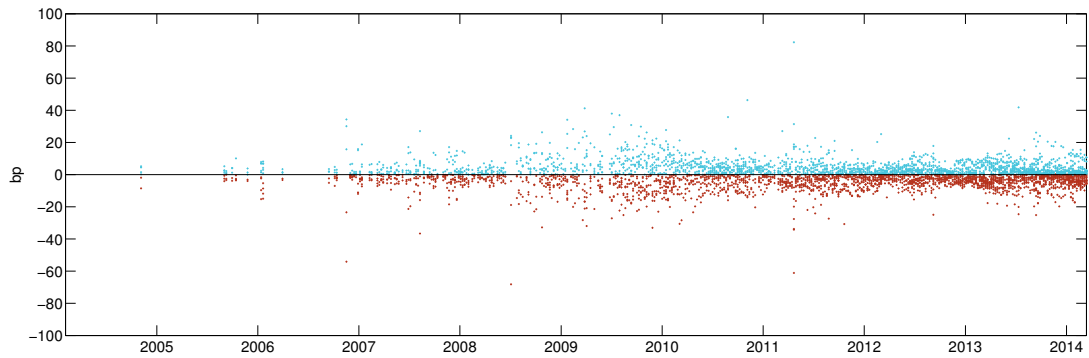
Table 5: We measure the “fit” of our yield curve models to market data, as well as the cross-market pricing of Islamic issues using the conventional yield curve by comparing the market yield with model yield (2004–2014). Excess market yield is the market yield less the arbitrage-free yield. Underpricing measures occurrence of positive excess market yields. Only yield curves based on at least 5 observations and minimum 5 years between the longest and shortest maturity are evaluated. Weeks rejected are the number of weeks where the null hypothesis (that average market yield is equal to average arbitrage-free yield) is rejected at 10% and 5% significance levels, respectively. N refers to the number of price observations.

			Excess market yield (%)				Under/Overpriced (%)		Weeks rejected	
N Valid Weeks			Mean	S.D.	Min	Max	Under	Over	10%	5%
Diebold-Li										
Conventional	12512	501	-0.004	0.123	-2.066	2.321	48.59	51.41	1	0
Islamic	4905	361	-0.004	0.071	-0.682	0.823	48.03	51.97	3	1
Nelson-Siegel										
Conventional	12512	501	-0.002	0.103	-1.350	2.198	48.79	51.21	3	1
Islamic	4905	361	-0.001	0.057	-0.735	0.507	49.24	50.76	2	2
Cross-Market										
Nelson-Siegel	5335	498	0.050	0.118	-2.798	1.054	75.73	24.27	247	222
Diebold-Li	5335	498	0.040	0.134	-2.828	0.979	67.07	32.93	202	170

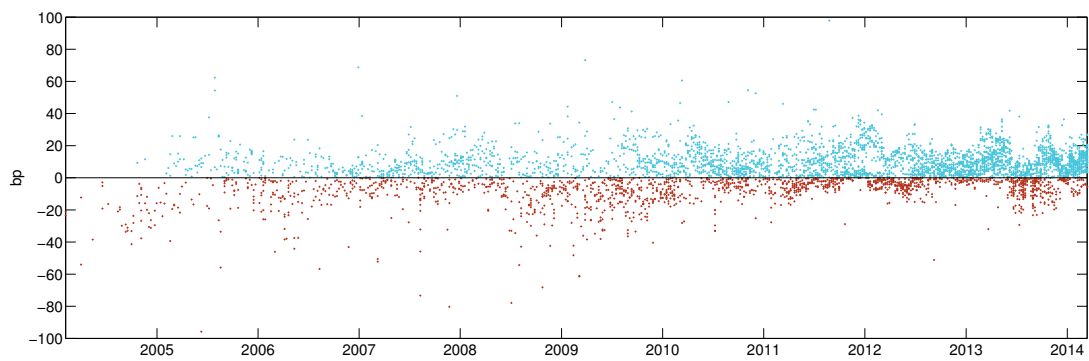
Cross-market has fewer valid weeks than the conventional curves because 3 conventional weeks completely lack Islamic trades, but larger N than the Islamic curves because more observations can be priced with valid conventional curves.



(a) Conventional (Diebold-Li)



(b) Islamic (Diebold-Li)



(c) Cross-Market: Islamic using Conventional (Diebold-Li)

Figure 6: Week-by-week spread (in basis points) between market and arbitrage-free yield using Diebold-Li zero curves, for conventional, Islamic and cross-market (Islamic issues priced with conventional zero curve) issues, 2004–2014. Each point represents the yield spread of a specific bond issue at a point in time. A positive yield (blue) suggests underpricing, a negative (red) overpricing. Excludes weeks where the curve was constructed from less than five observations, or there was less than five years between the shortest and longest maturity.

6.3 Arbitrage-free yields

Having established that both Nelson-Siegel and Diebold-Li models produce generally unbiased fits, we derive the arbitrage-free price of a bond with a given set of cash flows traded at the conventional market. We restrict this method with our criterion of excluding yield curves based on less than five observations or less than five years between the longest and shortest maturity. However, the availability of conventional data means that we can effectively price almost every Islamic issue between 2004 and 2014.

Some aggregate results of this approach have already been shown in section 6.2, specifically in table 5 and fig. 6c. Descriptives for the full sample of 5335 price observations together with divisions into maturity buckets are shown in table 6 computed with the Diebold-Li model. Corresponding statistics computed using Nelson-Siegel curves are found in appendix B. Statistics are shown both for the entire time period and for 2007–2014. Note that a single issue will appear multiple times and in multiple buckets as it changes throughout its lifetime.

Again, the descriptives suggest a pattern of systematic price deviations for Islamic bonds compared to conventional issues. Notably, the mean yield is generally higher, by as much as 10 basis points for 1–3 year maturities.

Our sample includes discount, coupon and Murabaha sukuk. During transitional periods, two different types of bonds are traded simultaneously on the secondary market. The descriptives for these classes are also shown in table 6. While comparing the differences between different kinds of sukuk is beyond the scope of this paper, there is some evidence that the most recent “version” of sukuk introduced in 2013, Murabaha, suffers from greater mispricing than coupon-bearing sukuk based on the Bai Bithaman-Ajil principles.

Looking again at fig. 6c, there appears to be a change to the mispricings over time at the issue-specific level. The incidence of possible overpriced outliers appears to sharply decrease after 2010 resulting in a strong bias towards underpricing that continues into 2014 with “clusters” of overpricing in 2012 and mid-2013.

Results from difference tests of arbitrage-free yields are shown in table 6. The results are highly significant and consistent with findings from comparing yield curves for the respective markets. Short-to-middle term sukuk suffer from greater mispricing than longer term issues. The yield spread is significantly negative or insignificantly different from zero for maturities beyond 7 years (there is similar convergence in the Nelson-Siegel model; the yield spread is still significantly above zero).

All in all, there is strong evidence against the hypothesis that investing in sukuk is categorically associated with equal, or lower, returns to investing in conventional bonds. While the differences are small in an absolute sense, this is in principle after having controlled for differences in maturity and coupon rates on securities with identical counter party risk.

Table 6: Islamic market and arbitrage-free yields as computed by the conventional Diebold-Li yield curve. Only arbitrage-free yields computed from curves with at least five observations and at least five years between the longest and shortest maturity are included. Panels A-D shows the sample for 2004–2014 (entire period) and 2007–2014 (restricted). Sukuk type panels show the respective “lifetime” periods. Yields are expressed on a semi-annual compounding basis. *N* refers to the number of price observations.

	N	Market yield (%)		Model yield (%)		Difference (%-units)		T-test		
		Mean	S.D.	Mean	S.D.	Mean	S.D.	t	1-sided p	2-sided p
Entire sample										
2004–2014	5335	3.401	0.461	3.361	0.476	0.040	0.134	21.656	0.000***	0.000***
2007–2014	4874	3.394	0.450	3.347	0.464	0.047	0.124	26.431	0.000***	0.000***
Panel A: Very short (<1 year)										
2004–2014	1454	3.025	0.403	2.984	0.382	0.041	0.171	9.086	0.000***	0.000***
2007–2014	1233	3.003	0.383	2.954	0.362	0.049	0.158	10.972	0.000***	0.000***
Panel B: Short maturity (1–3 years)										
2004–2014	1064	3.196	0.345	3.095	0.332	0.100	0.130	25.107	0.000***	0.000***
2007–2014	988	3.164	0.308	3.055	0.280	0.109	0.115	29.749	0.000***	0.000***
Panel C: Medium maturity (3–7 years)										
2004–2014	1605	3.519	0.242	3.485	0.266	0.034	0.107	12.814	0.000***	0.000***
2007–2014	1503	3.505	0.229	3.461	0.240	0.044	0.097	17.672	0.000***	0.000***
Panel D: Long maturity (7+)										
2004–2014	1212	3.877	0.329	3.884	0.324	-0.007	0.095	-2.436	0.993	0.015*
2007–2014	1150	3.864	0.325	3.869	0.319	-0.005	0.095	-1.780	0.962	0.075
Panel I: Discount GII										
2004–2011	126	3.344	0.470	3.383	0.488	-0.040	0.204	-2.188	0.031*	0.985
Panel II: Coupon-based GII										
2005–2014	4032	3.483	0.419	3.440	0.447	0.044	0.125	22.207	0.000***	0.000***
Panel III: Murabaha GII										
2013–2014	114	3.932	0.465	3.876	0.447	0.055	0.101	5.881	0.000***	0.000***

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

6.4 Liquidity regression results

The results of the yield curve and issue-specific regressions on liquidity are presented in table 8 and table 9, respectively.

As described above, we only have (nearly) week-to-week data without gaps for the time period 2009–2014. Thus, in order to estimate the first differenced regression, eq. (5.10), we restrict the time period. In the eight cases where the Islamic week fails the criterion of having at least five observations or five years between the longest and shortest maturity we simply impute the Islamic zero rates.

The results of estimating eq. (5.10) are rather disappointing (see table 8). Only in the case of the ten year spread is the liquidity measure even significant, and even in that case, the R^2 is still very low, showing that the average bid-ask spread has practically zero explanatory power on the spread between Islamic and conventional interest rates. The results are in line with previous studies on markets with thin trading data where other measures generally show higher significance.

The pooled cross-sectional regressions, as outlined in eq. (5.12) and presented in table 9, suggest that our measures of issue-specific liquidity could possibly have some explanatory power. Estimated as univariate models (I), they are not significant and have virtually nil explanatory power measured by R^2 . By controlling for bond-specific factors, the measures do become significant (II), and furthermore adding time dummies increases the explanatory power further (III). While both measures are expected to have positive signs (more illiquid instruments should be priced further away from the arbitrage-free price) bid-ask is negative and the incidence of zero returns changes sign twice.

The high significance of the time dummies (and their substantial contribution to R^2) seems to agree with the notion that mispricings occur in “clusters” and that controlling for time over cross-sections is important. Furthermore, the fact that including bond-specific characteristics increases the significance of both liquidity measures suggests that they might be negatively confounded with these variables.

From the earlier sections, it is natural to consider time to maturity significant, since we have seen strong evidence that the yield spread diminishes for long-term bonds (consistent with its sign). Liquidity is naturally negatively confounded with increasing maturity, since a bond is nothing more than the promise of future money. Including maturity increases the significance of the liquidity measures.

With these results in mind, we reestimate eq. (5.12), controlling more precisely for time (using quarterly dummies). We also include a “Murabaha dummy” since we have seen some evidence that such issues are more mispriced than earlier GII issues (discount GII are implicitly controlled for by the coupon rate).

The results are shown in table 10. We estimate the regression both for all Islamic issues (MITB and GII), as well as for GII only. We do not show the values of the time dummies, as the magnitude

of their coefficients is not interesting in itself; virtually all are highly significant.

Controlling for time more precisely further raises the explanatory power; these variations likely reflect macroeconomic and “sentiment” factors in the market. Notably, by increasing the amount of time dummies, the significance of the bid–ask spread disappears. Given its non-intuitive negative sign in the previous regressions – suggesting that a wider bid-ask spread leads to a smaller yield spread – this is a good result. Like other authors (e.g. Chen et al. (2007)), we find that the bid-ask spread is not a very useful measure. Emerging markets, such as Malaysia, generally have poor data. In such markets the bid-ask spread is notorious for consequently misestimating liquidity (Bekaert et al., 2007).

However, the proportion of zero returns remains significant even after controlling more precisely for time (and Murabaha). This suggests that transitory illiquidity can have explanatory power on the mispricing of sukuk relative to a conventional “counter-part”. The Murabaha dummy is significant at the 5% level only when comparing GII, suggesting a small but likely insignificant incremental mispricing of these issues. The significance of these results should not be overstated as it is likely that the regressions still suffer from omitted variable bias and potential endogeneity issues.

Table 7: Descriptives of liquidity measures used in liquidity regressions. % Zeros is the proportion of zero returns per week. Bid-Ask is the basis point spread between bid and ask yield. The average bid-ask is the average of end-of-week basis point bid-ask spreads in a given week on the respective market (only Islamic, only conventional and aggregate).

	N	Average	Min	Max
Issue-specific				
% Zeros	5335	0.46	0.00	1.00
Bid-Ask	2281	4.64	0.00	185.87
Average Bid-Ask				
Islamic	261	4.32	1.45	14.94
Conventional	263	6.85	3.56	10.89
Market	263	6.02	3.15	9.95

Table 8: Estimation of $\Delta \text{Yield Spread}_{t,t-1} = \beta_1 \Delta \text{Liquidity}_{t,t-1} + u_t$ for six zero rates (Diebold-Li curve) and three different market liquidities (only Islamic, only Conventional and Market-wide), 2009–2014. Each row thus shows three univariate models for a single maturity.

Spread	Islamic	Conventional	Market
1YR	1.593 (0.900)	2.472 (1.669)	3.595 (1.844)
R^2	0.012	0.008	0.014
2YR	0.037 (0.364)	0.364 (0.677)	0.334 (0.751)
R^2	0.000	0.001	0.001
3YR	-0.415 (0.228)	-0.383 (0.443)	-0.698 (0.490)
R^2	0.013	0.003	0.008
5YR	-0.277 (0.169)	-0.454 (0.347)	-0.511 (0.385)
R^2	0.007	0.007	0.007
7YR	0.214 (0.179)	-0.110 (0.363)	0.251 (0.403)
R^2	0.005	0.000	0.001
10YR	0.714** (0.270)	0.331 (0.524)	1.15* (0.577)
R^2	0.026	0.002	0.015

$N = 262$ for market and conventional, $N = 259$ for Islamic (no bid-ask quotes were available in two weeks). Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 9: Yield spread (Islamic market yield less arbitrage-free yield, in basis points) determinants based on issue-specific factors (annual coupon rate (in %); time to maturity (in years)), the slope of the conventional yield curve (difference between 10- and 2-year zero rate (basis points)) and two different, issue-specific liquidity measures (cf. Chen et al. (2007)): The bid–ask spread (in basis points) and the proportion of zero returns per week. Based on weekly samples, 2004–2014, of 61 GII and 164 MITB issues.

	Bid-Ask (I)	Bid-Ask (II)	Bid-Ask (III)	% Zeros (I)	% Zeros (II)	% Zeros (III)
Bid–Ask	−0.060 (0.15)	−0.236* (0.11)	−0.238* (0.11)			
% Zeros				0.510 (0.52)	−1.354** (0.51)	1.195* (0.49)
Coupon		−7.063 (22.07)	0.663 (29.23)		145.118*** (17.31)	109.136*** (16.55)
Maturity		−1.101*** (0.10)	−1.103*** (0.10)		−1.052*** (0.08)	−1.004*** (0.07)
Slope		−3.244*** (0.79)	−1.104 (1.31)		−3.431*** (0.59)	−3.373*** (0.86)
2005			26.536*** (2.81)			23.407*** (5.31)
2006			10.579 (5.69)			14.140** (5.36)
2007			24.411*** (3.27)			18.467*** (5.39)
2008			28.377*** (2.94)			18.486*** (5.32)
2009			24.317*** (2.53)			19.670*** (5.14)
2010			27.925*** (2.56)			24.791*** (5.17)
2011			25.785*** (2.62)			25.354*** (5.22)
2012			26.187*** (2.74)			24.048*** (5.24)
2013			27.134*** (2.57)			25.961*** (5.21)
2014			29.196*** (2.35)			29.290*** (5.14)
Constant	3.553*** (0.68)	12.866*** (1.14)	−15.503*** (3.50)	3.755*** (0.25)	7.953*** (0.70)	−15.832** (5.44)
Observations	2281	2281	2281	5335	5335	5335
Adjusted R^2	0.001	0.116	0.229	−0.000	0.062	0.149

Robust standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 10: Reestimation of eq. (5.12) (see table 9 for variable specification) using 40 quarterly time dummies (Q12004–Q12014, not shown) and Murabaha dummy (1 if the bond is issued under Murabaha principles). Shows results using all Islamic issues (61 GII and 164 MITB), as well as GII only.

	GII and MITB		GII Only	
	Bid-Ask	% Zeros	Bid-Ask	% Zeros
Bid-Ask	−0.218 (0.11)		0.152 (0.14)	
% Zeros		1.522** (0.48)		3.694*** (0.56)
Murabaha	−1.042 (1.23)	1.202 (1.24)	−1.559 (1.30)	3.381* (1.33)
Coupon	−11.908 (37.22)	107.376*** (16.47)	−204.101 (117.28)	−47.000 (51.74)
Maturity	−1.112*** (0.10)	−1.035*** (0.07)	−0.929*** (0.11)	−0.962*** (0.08)
Slope	−1.322 (2.76)	−1.989 (1.88)	−1.331 (2.87)	−2.031 (1.87)
Constant	−15.843** (5.20)	−18.818*** (3.36)	−20.871*** (5.30)	−20.561*** (3.33)
Observations	2281	5335	1838	4272
Adjusted R^2	0.283	0.193	0.267	0.200

Robust standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

7 Conclusion

The dual capital market system in Malaysia constitutes an interesting study prospect. The establishment of an Islamic financial market could potentially increase financial inclusion. The interaction between the two markets are, however, subject to market frictions creating deviations from the no-arbitrage condition. The existence of such frictions creates obstacles for increased participation and market growth.

This study has found a persistent spread disproving the hypothesis that yields on sukuk are equal to or lower than those of conventional bonds, in line with previous studies (Safari et al., 2013; Fathurahman & Fitriati, 2013). We have found that the yield of a sukuk over a conventional bond is significantly different from zero and in general positive. This applies both when comparing zero yield curves as well as when considering price deviations at the issue specific level. The yield spread quantifies the cost of frictions associated with the government sukuk market.

We have found that during the time period studied, the spread on zero rates has in general reverted towards a persistent level with temporary fluctuations. We have argued that these fluctuations can, at least in part, be explained by illiquidity issues on the Islamic market. We have found no evidence at an aggregate level when measuring liquidity using the average bid-ask spread. At the issue-specific level, the bid-ask spread also yielded no significant results.

The percentage of zero-returns are arguably a better proxy for liquidity on markets with meager data and better captures the illiquidity effect in our model. At the issue-specific level the regression on zero returns showed significance when controlling for other bond characteristics and time.

The positive influence of time dummies suggests the mispricings occurs in clusters. Many events are likely to have an effect on the yield spread, however, the detailed impact of such events is beyond the scope of this study.

The most important result of this study is the existence of a persistent spread between the two markets that is not supported by differences in fundamental value. We have found some support that it can be partly explained by illiquidity issues. However, the driving factors of such illiquidity are yet to be fully understood.

8 Limitations and suggestions for future research

8.1 Limitations

Bonds are notoriously thinly traded. The Malaysian bond market is no exception and the sukuk market has very poor trading data. Bloomberg, accounts for approximately 91.8% of our data. As data was missing we complemented it with data from BNM and Bond Info. It is possible that this introduces biases in the data.

The most glaring data issue is the unavailability of daily quotes. As a result, we use “end-of-week” yields which might not reflect the “true” yield on the asset. With this in mind, underpricing of Islamic issues is still (very) prevalent in 2013 and 2014—the periods with the highest quality of data—suggesting that this is not a major source of bias. Running tests using only Friday-quotes does not qualitatively change the results but tends to suggest larger deviations (not shown). We argue that carrying forward is defensible, since the information used is the most up-to-date on the market in a given week, however, apparently not valuable enough for the market to adjust its prices. The results still suggest that, at the end of each week, Islamic issues tend to be underpriced using the most recently available information. The study is limited in time to 2004–2014 and in practice often even shorter time periods due to the lack of data which limits its applicability.

The study fulfills its purpose as an explorative study of an emerging market. However, the models might fail to capture important elements. Notably, a preferred approach to estimating the illiquidity component at the aggregate level would be by constructing a vector autoregression (VAR) model. A properly constructed VAR model should capture important elements of causality. In this paper it is assumed that causality goes from the conventional to the Islamic market. Our linear regressions likely do not properly control for all omitted variables, and their results should not be overstated. Our comparisons are made on the basis of a three factor model of the yield curve estimated using thin data and sometimes unsatisfactory procedures. The resulting yield curves’ inability to model the true dynamics of the bond market can both exaggerate and undermine our results.

8.2 Suggestions for future research

What drives demand and who is investing in sukuk is currently not established. This paper assumes the presence of arbitrageurs that are indifferent between the products on the respective markets. This assumption is firmly established in theory but not verified in practice. Behavioural aspects are likely to have an effect on the pricing of sukuk. Some aspects of the primary market are still ambiguous. The implications of introducing Murabaha are not yet determined. Lastly, the derivatives market of sukuk is today very limited and taking a short position is not possible. Investigating the consequences of this void could constitute interesting research.

References

- Azmat, S., Skully, M., & Brown, K. (2013). The Shariah compliance challenge in Islamic bond markets. *Pacific-Basin Finance Journal*. Article in press.
- Bacha, O. I. & Mirakhor, A. (2013). *Islamic Capital Markets: A Comparative Approach*. Somerset, NJ, USA: Wiley.
- Bank Negara Malaysia (2001). Financial Sector Master Plan 2001–2010. Available online: http://www.bnm.gov.my/index.php?ch=en_fsmp&pg=en_fsmp_book&ac=24&lang=en.
- Bank Negara Malaysia (2005). Guidelines on regulated short selling of securities in the wholesale money market. Published on 7th of October, 2005. Available online: https://fast.bnm.gov.my/fastweb/public/files/Reg_ShortSell_Oct2005.pdf.
- Bank Negara Malaysia (2011). Financial Sector Blueprint 2011–2020. Published on 21 December 2011. Available online: http://www.bnm.gov.my/files/publication/fsbp/en/BNM_FSBP_FULL_en.pdf.
- Bank Negara Malaysia (2012). Frequently Asked Question: Government Investment Issue. Available online: [http://iimm.bnm.gov.my/view.php?id=94&dbIndex=0&website_id=14&ex=1381675757&md=%98%BDT!%08%D4%AAEU2J4\)%F7%F2%D0](http://iimm.bnm.gov.my/view.php?id=94&dbIndex=0&website_id=14&ex=1381675757&md=%98%BDT!%08%D4%AAEU2J4)%F7%F2%D0).
- Bank Negara Malaysia (2013). Information Note: Government Investment Issue. Available online: http://iimm.bnm.gov.my/view.php?id=95&dbIndex=0&website_id=14&ex=1395748510&md=%14Z2B%CF%9C%1C%26%C3V950%22%BAC.
- Bank Negara Malaysia (2014). Bank Negara Malaysia website. <http://http://bnm.gov.my>.
- Bank Negara Malaysia & Securities Commission Malaysia (2009). Malaysian debt securities and sukuk market: A guide for issuers and investors. Available online: <http://www.sc.com.my>. Guidebook.
- Bao, J., Pan, J., & Wang, J. (2011). The illiquidity of corporate bonds. *The Journal of Finance*, 66(3), 911–946.
- Bekaert, G., Harvey, C. R., & Lundblad, C. (2007). Liquidity and expected returns: Lessons from emerging markets. *Review of Financial Studies*, 20(6), 1783–1831.
- BNM Bond Info (2014). Bond Info Hub website. <http://http://bondinfo.bnm.gov.my>.
- Cakir, S. & Raei, F. (2007). Sukuk vs. eurobonds: Is there a difference in Value-at-Risk? *IMF Working Paper*. October 2007. WP/07/237.
- Chen, L., Lesmond, D. A., & Wei, J. (2007). Corporate yield spreads and bond liquidity. *The Journal of Finance*, 62(1), 119–149.
- Coroneo, L., Nyholm, K., & Vidova-Koleva, R. (2008). How arbitrage-free is the Nelson-Siegel model? *ECB Working Paper Series*, (874). February 2008.

- Diebold, F. X. & Li, C. (2006). Forecasting the term structure of government bond yields. *Journal of Econometrics*, 130, 337–364.
- Fathurahman, H. & Fitriati, R. (2013). Comparative analysis of return on sukuk and conventional bonds. *The American Journal of Economics*, 3(3), 159–163.
- Fleming, M. J. (2003). Measuring treasury market liquidity. *Economic Policy Review*, Federal Reserve Bank of New York, September 2003, 83–108.
- Godlewskic, C. J., Turk-Arissb, R., & Weilla, L. (2013). Sukuk vs. conventional bonds: A stock market perspective. *The Journal of Comparative Economics*, 41(3), 745–761.
- Hagan, P. S. & West, G. (2008). Methods for constructing a yield curve. *WILMOTT magazine*, May/June, 78–81.
- Iqbal, Z. & Mirakhor, A. (2013). *Economic Development and Islamic Finance*. Washington, DC: Directions in Development: World Bank.
- Jobst, A. A. (2007). The economics of Islamic finance and securitization. *The Journal of Structured Finance*, 13(1), 6–27.
- Kim, D. H. & Orphanides, A. (2007). The bond market term premium: what is it, and how can we measure it? *BIS Quarterly Review*, June 2007.
- Krichene, N. (2013). *Islamic Capital Markets: Theory and Practice*. John Wiley & Sons Singapore Pte. Ltd.
- Lesmond, D. A., Ogden, J. P., & Trcinka, C. A. (1999). A new estimate of transaction costs. *The Review of Financial Studies*, 12(5), 1113–1141.
- McDonald, R. L. (2006). *Derivatives Markets*. Pearson Education, Inc., second edition.
- Mosaid, F. E. & Boutti, R. (2014). Sukuk and bond performance in malaysia. *International Journal of Economics and Finance*, 6(2).
- Nelson, C. R. & Siegel, A. F. (1987). Parsimonious modelling of yield curves. *Journal of Business*, 60(4), 473–489.
- Omar, M. A., Abduh, M., & Sukmana, R. (2013). *Fundamentals of Islamic Money and Capital Markets*. John Wiley & Sons Singapore Pte. Ltd.
- Ramasamy, R., Munisamy, S., & Helmi, M. H. M. (2011). Relative risk of Islamic sukuk over government and conventional bonds. *Global Journal of Management and Business Research*, 11(6), 4–12.
- Rosly, S. A. & Sanusi, M. M. (1999). The application of Bay' Al-'Inah and Bay' Al-Dayn in malaysian Islamic bonds: An Islamic analysis. *International Journal of Islamic Financial Services*, 1(2).
- Safari, M., Ariff, M., & Mohamad, S. (2013). Do debt markets price sukuk and conventional bonds differently? *Journal of King Abdulaziz University Islamic Economies*, 26(2), 113–149.

Svensson, L. E. (1995). Estimating forward interest rates with the extended Nelson & Siegel method. *Sveriges Riksbank Quarterly Review*, 3(1), 13–26.

Tsay, R. S. (2005). *Analysis of Financial Time Series*. Hoboken, New Jersey.: John Wiley & Sons, 2nd edition.

Veronesi, P. (2011). *Fixed Income Securities: Valuation, Risk, and Risk Management*. Hoboken, New Jersey: John Wiley & Sons, Inc., 1st edition.

Wooldridge, J. M. (2009). *Introductory Econometrics: A modern approach*. South-Western, 4th edition.

Appendices

Appendix A Sukuk principles

The below is based on information provided by Bank Negara Malaysia & Securities Commission Malaysia (2009).

Quard al-Hasan prohibits any official compensation for the lending of money. A sukuk based on this principle is issued and redeemed at par without any coupon payments. This principle is compliant with the strictest interpretation of Shariah. The first sukuk issued by the government of Malaysia was based on this principle. Quard al-Hasan sukuk are only used for short-term financing and only in moderation as it relies on the benevolence of the investor. They are not traded on the secondary market.

Bai al-Inah sukuk are asset-based and yield a profit. The issuer sells the investor an asset for a direct cash payment and then the investor directly sells back the asset but on a pre-agreed deferred payment basis. Sukuk solely based on Bai al-Inah have no coupon payments. The sukuk is asset-based but not asset-backed as the asset does not constitute collateral.

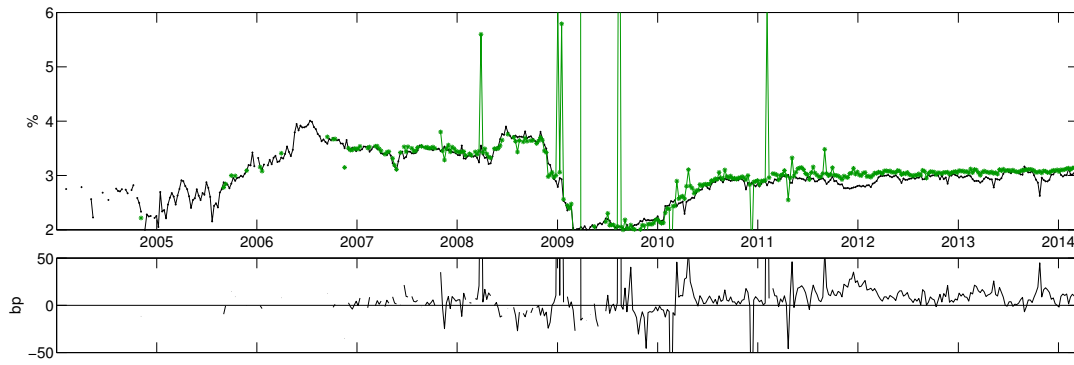
Bai Bithaman Ajil allows for purchases on a deferred payment and instalment basis. A sukuk based on the Bai Bithaman Ajil principle in conjunction with the Bai al-Inah principle is coupon-bearing as it allows for instalments.

Murabaha sukuk are also asset-based but do not yield a profit like Bai al-Inah. Murabaha sukuk are often based on the transaction of a commodity and the cash flows are set on a cost plus sale basis instead of being profit-based. The transaction structure is more complex and engages more parties; in addition to the issuer and the investor it includes a *Special Purpose Vehicle* (SPV) and the commodity market or corresponding asset market.

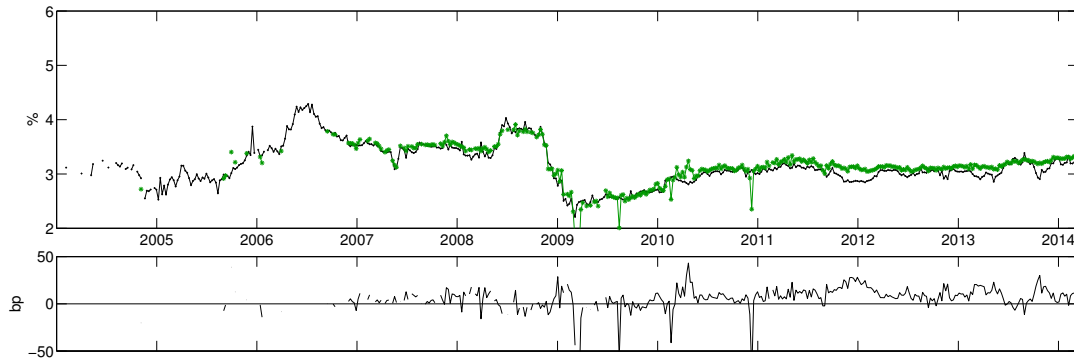
An investor appoints an agent, the SPV, to purchase the asset on their behalf. The SPV sells the purchased asset to the government at a mark-up price to be paid on a deferred payment and instalment basis. The sukuk issuer sells the assets to raise the funds.

Bay al-Dayn is simply the process of selling debt. The principle is the foundation of the secondary bond market. Bay al-Dayn is considered Shariah compliant on the Malaysian market but not on all Islamic financial markets. Elements of Bay al-Dayn are present if a transferrable sukuk certificate is issued.

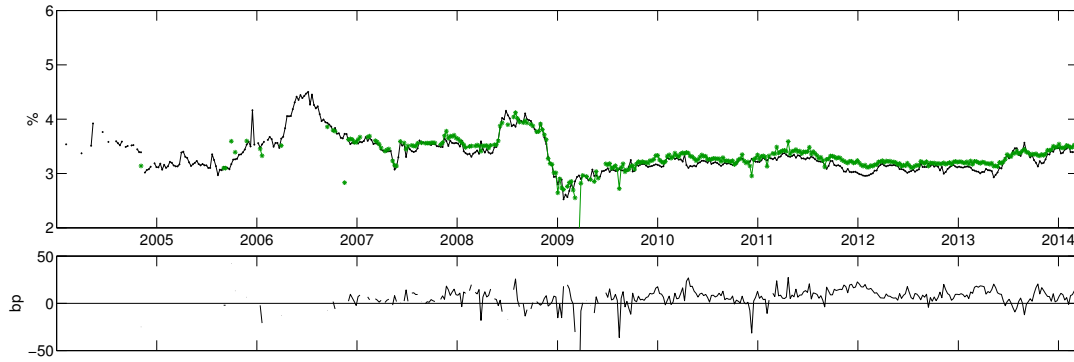
Appendix B Complementing tables and figures using Nelson-Siegel



(a) 1 year

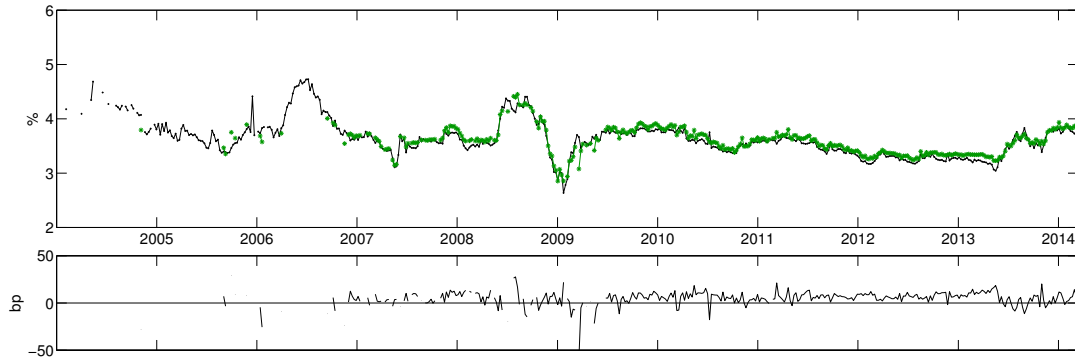


(b) 2 year

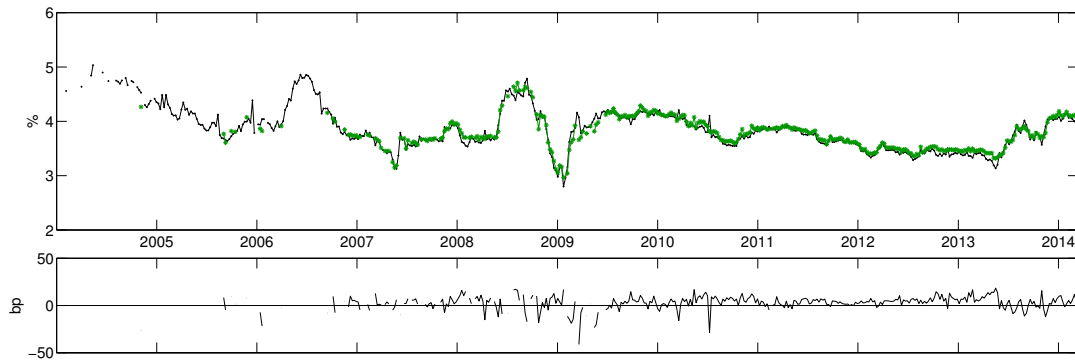


(c) 3 year

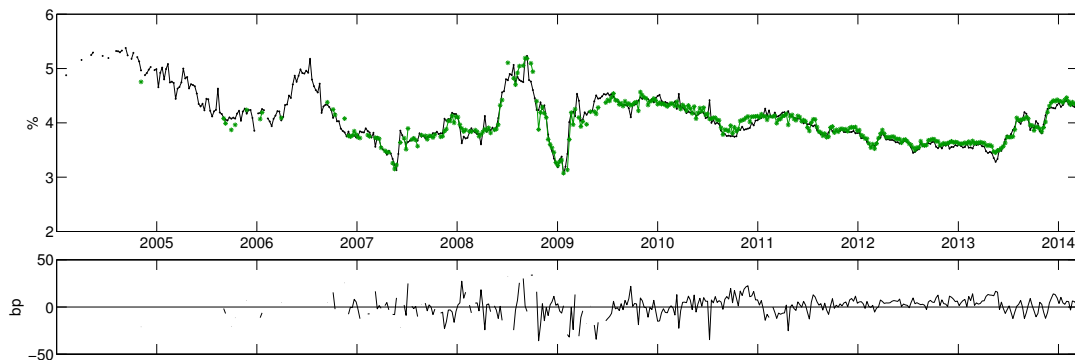
Figure 7: Time series of Nelson-Siegel yield curves at selected maturities. The upper panel shows Islamic (green) and conventional (black) on top of each other. The bottom panel shows the spread, Islamic less conventional, in basis points (bp). Only shows weeks with at least five observations and minimum five years between shortest and longest maturities (missing weeks are shown as gaps). *Continued on next page*



(d) 5 year

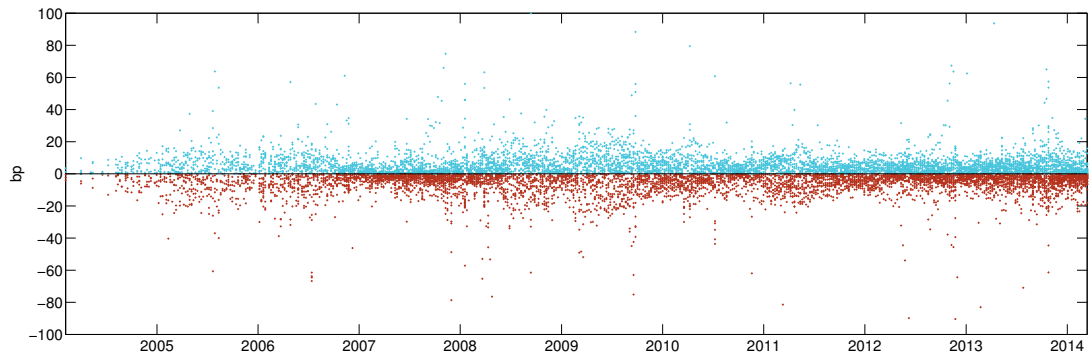


(e) 7 year

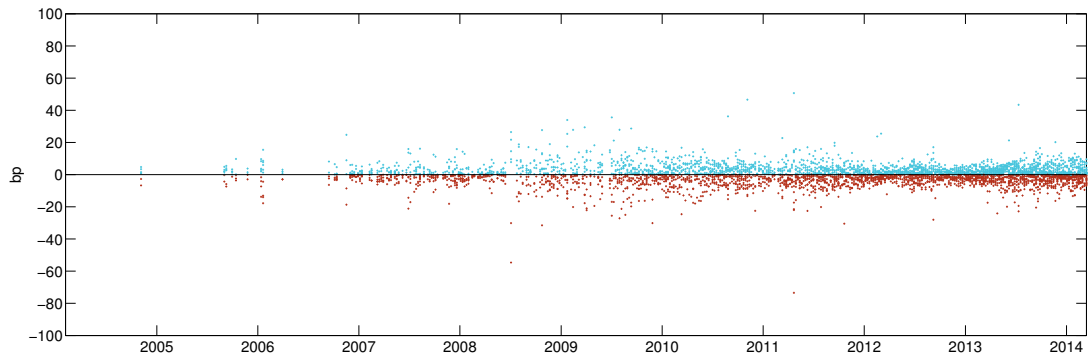


(f) 10 year

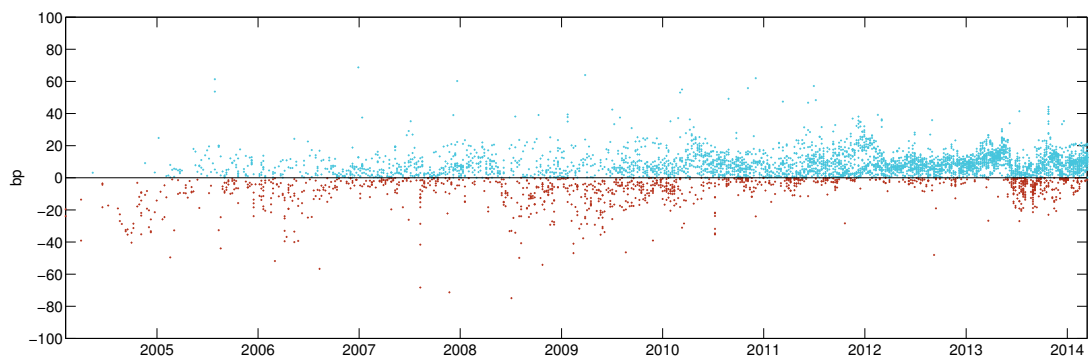
Figure 7: Continued Time series of Nelson-Siegel yield curves at selected maturities. The upper panel shows Islamic (green) and conventional (black) on top of each other. The bottom panel shows the spread, Islamic less conventional, in basis points (bp). Only shows weeks with at least five observations and minimum five years between shortest and longest maturities (missing weeks are shown as gaps).



(a) Conventional (Nelson-Siegel)



(b) Islamic (Nelson-Siegel)



(c) Cross-Market: Islamic using Conventional (Nelson-siegel)

Figure 8: Week-by-week spread (in basis points) between market and arbitrage-free yield using Nelson-Siegel zero curves, for conventional, Islamic and cross-market (Islamic issues priced with conventional zero curve) issues, 2004–2014. Each point represents the yield spread of a specific bond issue at a point in time. A positive yield (blue) suggests underpricing, a negative (red) overpricing. Excludes weeks where the curve was constructed from less than five observations, or there was less than five years between the shortest and longest maturity.

Table 11: Islamic market and arbitrage-free yields as computed by the conventional Nelson-Siegel yield curve. Only arbitrage-free yields computed from curves with at least five observations and at least five years between the longest and shortest maturity are included. Panels A-D shows the sample for 2004–2014 (entire period) and 2007–2014 (restricted). Sukuk type panels show the respective “lifetime” periods. Yields are expressed on a semi-annual compounding basis. *N* refers to the number of price observations.

	N	Market yield (%)		Model yield (%)		Difference (%-units)		T-test		
		Mean	S.D.	Mean	S.D.	Mean	S.D.	t	1-sided p	2-sided p
Entire sample										
2004–2014	5335	3.401	0.461	3.351	0.467	0.050	0.118	30.872	0.000***	0.000***
2009–2014	4874	3.394	0.450	3.336	0.456	0.058	0.109	36.988	0.000***	0.000***
Panel A: Very short (<1 year)										
2004–2014	1454	3.025	0.403	2.984	0.388	0.041	0.151	10.301	0.000***	0.000***
2009–2014	1233	3.003	0.383	2.953	0.372	0.050	0.147	12.033	0.000***	0.000***
Panel B: Short maturity (1–3 years)										
2004–2014	1064	3.196	0.345	3.123	0.347	0.073	0.115	20.728	0.000***	0.000***
2009–2014	988	3.164	0.308	3.083	0.302	0.081	0.095	26.779	0.000***	0.000***
Panel C: Medium maturity (3–7 years)										
2004–2014	1605	3.519	0.242	3.463	0.262	0.056	0.097	23.133	0.000***	0.000***
2009–2014	1503	3.505	0.229	3.440	0.236	0.066	0.085	30.073	0.000***	0.000***
Panel D: Long maturity (7+)										
2004–2014	1212	3.877	0.329	3.845	0.341	0.032	0.094	11.934	0.000***	0.000***
2009–2014	1150	3.864	0.325	3.829	0.335	0.035	0.094	12.633	0.000***	0.000***
Panel I: Discount GII										
2004–2011	126	3.344	0.470	3.379	0.488	-0.035	0.196	-2.016	0.046*	0.977
Panel II: Coupon-based GII										
2005–2014	4032	3.483	0.419	3.426	0.437	0.057	0.111	32.737	0.000***	0.000***
Panel III: Murabaha GII										
2014–2014	114	3.932	0.465	3.868	0.432	0.064	0.009	7.184	0.000***	0.000***

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$